

Triple A

~ Advanced Amplicon Analysis ~

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User Manual

· From Launching to Reading Results ·

This manual covers the 36 graphical outputs generated by the platform, the exact format of the input files with reproducible examples, and the validated limitations of the current version.

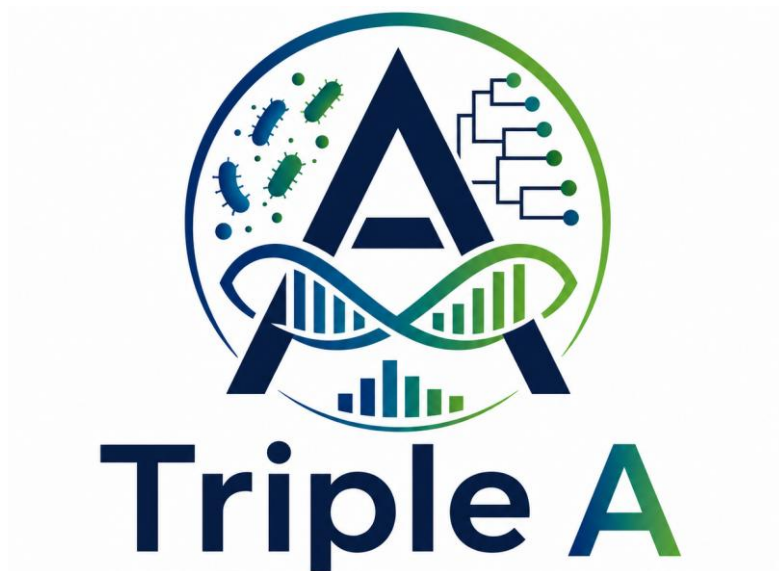


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1. What is Triple A?

The rapid expansion of high-throughput amplicon sequencing has enabled the characterization of increasingly complex microbial communities. However, transforming large abundance datasets into biologically meaningful conclusions remains challenging, as it requires the coordinated application of multiple statistical, ecological and functional analyses, often distributed across different software platforms. To address these limitations, we developed **Triple A (Advanced Amplicon Analysis)**, an integrated and reproducible framework that standardizes microbiome data analysis from data preprocessing to statistical inference and biological interpretation. Its primary objective is to transform raw microbial community abundance tables into robust statistical evidence and biologically meaningful conclusions through an integrated, reproducible workflow.

Unlike many bioinformatics tools that focus solely on generating figures or performing isolated statistical tests, Triple A was conceived as a complete analytical framework. It guides researchers through every stage of microbiome data analysis, from data import and quality validation to multivariate statistics, biomarker discovery and functional interpretation. The software organizes analyses according to the biological questions they answer rather than the statistical methods themselves, making it accessible to both experienced bioinformaticians and researchers with limited statistical training.

This **project** was developed around three fundamental principles:

- Standardization: analyses are performed using validated and reproducible workflows.
- Scientific rigor: integrating established methods widely accepted in microbiology.
- Biological interpretation: helping researchers understand what the statistical results actually mean in ecological and microbiological terms.

The following figure (Figure 1 & Figure 2) illustrates the complete workflow implemented in Triple A for the biological analysis of amplicon sequencing data, from data import and preprocessing to statistical inference, functional profiling and biological interpretation.

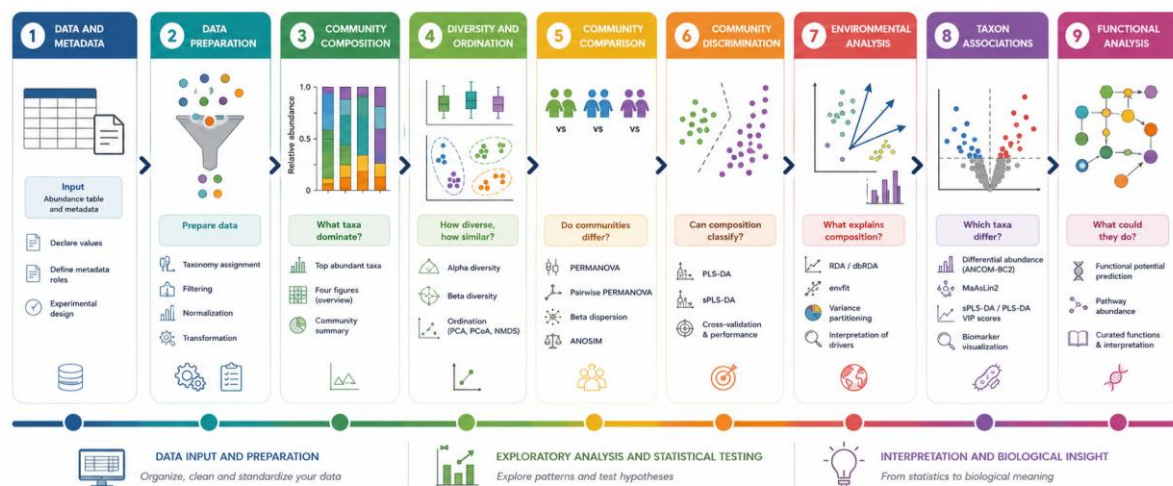


Figure 1. Graphical workflow followed in the analysis performed by the project Triple A

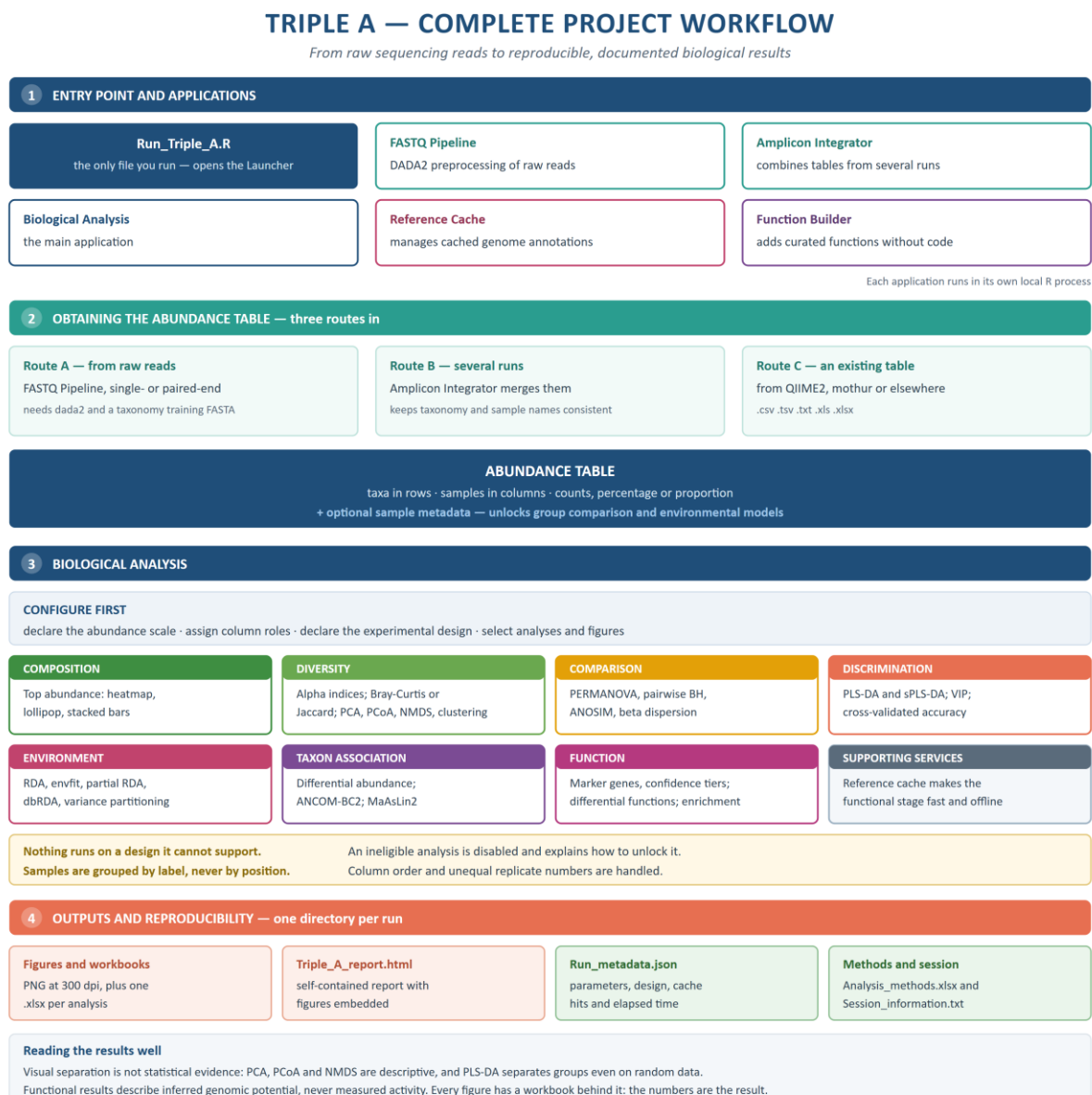


Figure 2. Complete Workflow followed in the analysis performed by the project Triple A.

2. Installation and first run

- Install R (4.4.x recommended). No other installation step is required.
- Unzip the distribution. A folder named Triple_A appears.
- Open and run Run_Triple_A.R. The Launcher opens in your browser.
- The Launcher checks required packages and offers to install anything missing. Most analyses need only common CRAN packages; ANCOM-BC2, MaAsLin2 and the FASTQ module need Bioconductor packages, offered on demand.

Keep the folder in a reasonably short path. Deep output paths can exceed the Windows 260-character limit and make a figure fail to save.

3. Project structure

Everything Triple A needs and everything it produces lives inside the project folder. Moving or copying that single directory moves the whole working environment, including the reference cache and every past result.

Table 1. Main project folders

Location	Responsibility
Run_Triple_A.R	The only file you run. It opens the Launcher, from which every application is started.
AAApp/	The Launcher and the five applications. Each subfolder is one application.
AAApp/Common/Engine/	The analysis engine: all the science, independent of any interface. Triple_A.R exposes the public API, Core/ holds the analysis modules and Extensions/ the biological function registry.
Plugins/	The seven analysis plugins, discovered at runtime. Each is registered with a uniform contract, so a new method can be added without modifying the engine.
Resources/Documentation/	The user guides, also shown by the Help button inside each application.
Resources/FunctionalDB/	The curated functional database, plus any function you add with the Function Builder.
Cache/	The reference-genome cache: a SQLite index and the GFF annotations already resolved from NCBI.
Results/	Everything a run produces. Nothing is written anywhere else.

Each run creates one directory under Results/Runs/, named with its timestamp, containing one subdirectory per analysis plus the run's own reproducibility record. The metadata files sit at the root of the run directory, not in a subfolder.

Table 2. Anatomy of a single run

Item	Contents
<Analysis_name>/	One folder per analysis, holding its figures as PNG at 300 dpi and one Excel workbook* with every number behind them.
Run_metadata.json	Run identifier, status, selected analyses, treatments, replicate design, cache hits and elapsed time.
Analysis_methods.xlsx	For each selected output: objective, data preparation, method, statistical test, significance criterion and limitations.
Session_information.txt	The R version and the exact version of every package used in the run.
Triple_A_manifest.xlsx	Every file the run produced.
Triple_A_report.html	A self-contained report gathering figures, tables and parameters, with the images embedded so the file can be moved or emailed. Optional PDF export through headless Chrome.

Because a run folder is complete in itself, it can be archived or shared as a unit and still document exactly how its results were produced. Results/Logs/ holds the startup messages of each application, and Results/FASTQ/ the output of the FASTQ module. Neither is tied to a particular run.

* No figure is the primary result. Each one is accompanied by the spreadsheet of the values used to draw it, in the same folder. Whenever a visual reading is uncertain, whether two groups really separate, whether a bar is meaningfully taller, the numbers in the workbook are the answer, and the picture is only a summary of them.

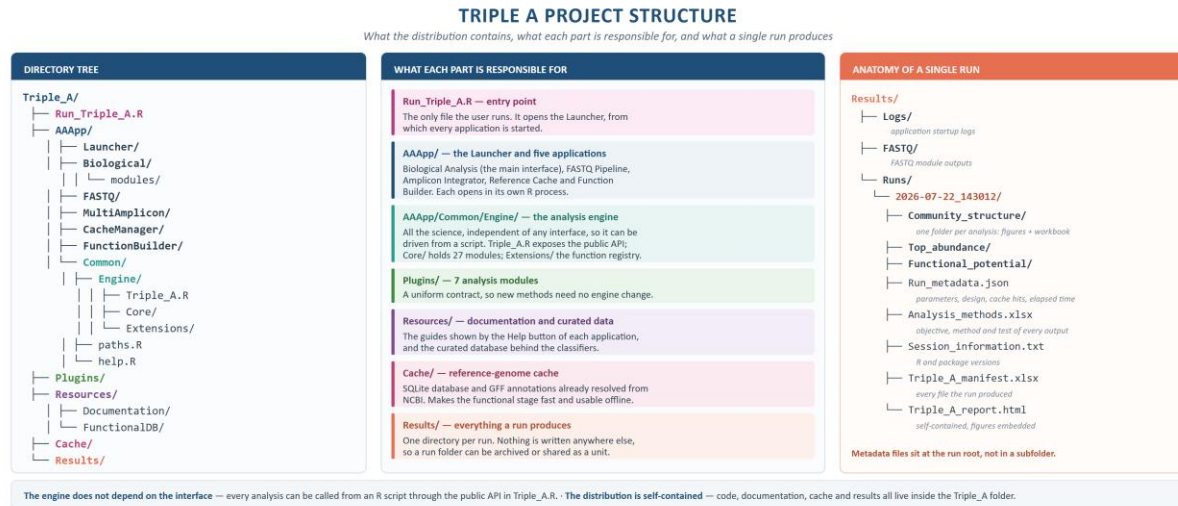


Figure 3. Detailed structure of the project and folders organization

4. Graphical user interface

Triple A has been designed as a guided analytical environment that organizes microbiome analyses according to the natural progression of a research project rather than as an isolated collection of statistical methods. The graphical user interface (GUI) follows a modular workflow that guides users from raw sequencing data to biological interpretation while minimizing manual preprocessing and reducing the likelihood of analytical errors.

The interface is divided into five major analytical modules (Figure 4), each corresponding to a different stage of the microbiome analysis workflow. Triple A is not a single window. It is a Launcher plus five applications, each opening in its own local R process. A failure in one module therefore cannot take the others down, and a long analysis does not freeze the interface you launched it from.

Every application has a Help button that shows the documentation for that module, read from Resources/Documentation/. Startup messages for each one are written to Results/Logs/.

Table 3. Major analytical modules

Module	Purpose
Launcher	Entry point. Checks required R packages and opens each application in its own R process.
FASTQ Pipeline	Processes raw reads with DADA2: quality assessment, trimming, ASV inference, taxonomy. Produces an abundance table.
Amplicon Integrator	Combines two or more compatible abundance tables into one consistent dataset, checking taxonomy and sample identifiers.
Biological Analysis	The main application: import, validation, experimental design, analysis selection, execution and result browsing.
Reference Annotation Cache	Inspects, imports, merges, verifies, cleans and exports the local SQLite/GFF reference cache.
Biological Function Builder	Adds new biological functions and pathways declaratively, without editing R code.

Analysis

**Biological Analysis (Triple A)**
Import abundance data and run taxonomic, statistical and functional analyses.
[Open Biological Analysis](#)

Data preparation

**Amplicon Integrator (Triple A)**
Validate and combine compatible amplicon count tables.
[Open MultiAmplicon](#)

**FASTQ Pipeline (Triple A)**
Run DADA2 preprocessing and export a compatible abundance table.
[Open FASTQ Pipeline](#)

Resources and extension

**Cache (Triple A)**
Inspect, merge, export and verify cached reference annotations.
[Open Cache Manager](#)

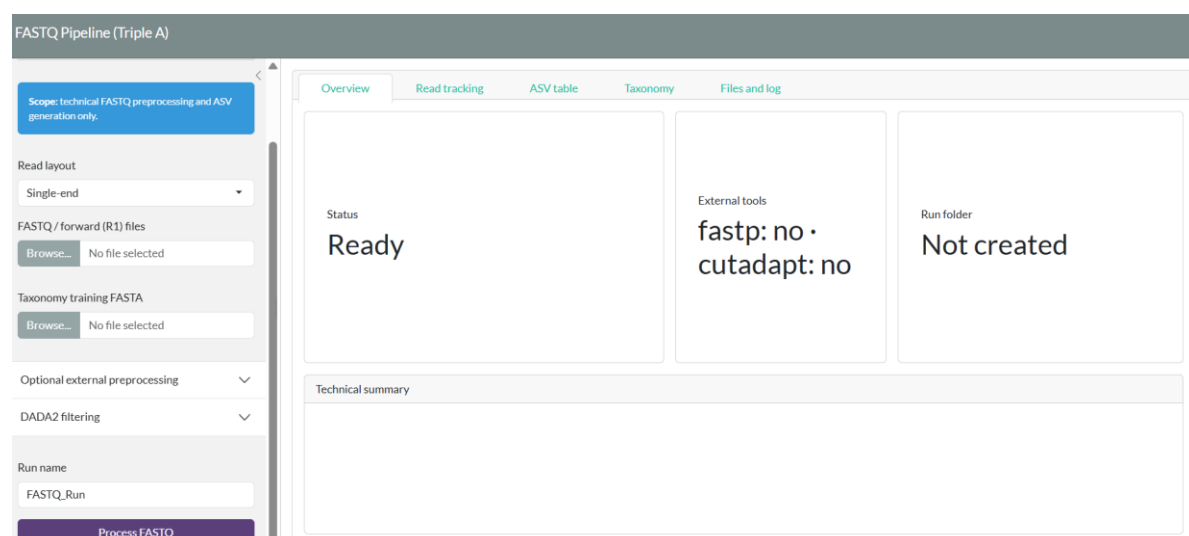
**Biological Function Builder (Triple A)**
Create and validate declarative biological functions without editing R scripts.
[Open Function Builder](#)

Figure 4. Launcher interface of Triple A

This organization enables users to perform the entire analysis within a single software environment, from sequencing reads to biological conclusions.

4.1. FASTQ module

The upstream route, for studies that start from raw reads. It runs the DADA2 workflow: optional primer removal, quality filtering and trimming, error-model learning, denoising, merging of paired reads, construction of the sequence table, chimera removal and taxonomic assignment. At the end of this module, Triple A produces a high-quality ASV abundance matrix together with its corresponding taxonomic annotations.



The screenshot shows the 'FASTQ Pipeline (Triple A)' interface. On the left is a sidebar with configuration options: 'Scope: technical FASTQ preprocessing and ASV generation only.', 'Read layout' (Single-end), 'FASTQ / forward (R1) files' (Browse... No file selected), 'Taxonomy training FASTA' (Browse... No file selected), 'Optional external preprocessing' (dropdown), 'DADA2 filtering' (dropdown), 'Run name' (FASTQ_Run), and a 'Process FASTQ' button. The main panel has tabs: Overview, Read tracking, ASV table, Taxonomy, and Files and log. The 'Overview' tab is active, showing a 'Status' box with 'Ready', 'External tools' with 'fastp: no' and 'cutadapt: no', and 'Run folder' with 'Not created'. Below these is a 'Technical summary' section.

Figure 5. FASTQ module interface of Triple A

4.2. Amplicon Integrator module

The Amplicon Integration module allows users to prepare abundance tables generated either within Triple A (FASTQ module) or by external pipelines such as QIIME2 or DADA2. This

module performs the integration of two or more abundance tables in a single dataset to analyse it in the biological analysis platform. The module performs:

- Taxonomy integration
- Validation of sample identifiers
- Abundance table verification
- Replicate integration
- Generation of an analysis-ready dataset.

This stage ensures that all downstream analyses are performed using standardized and internally consistent datasets. **Use it before the Biological Analysis module, never after.**

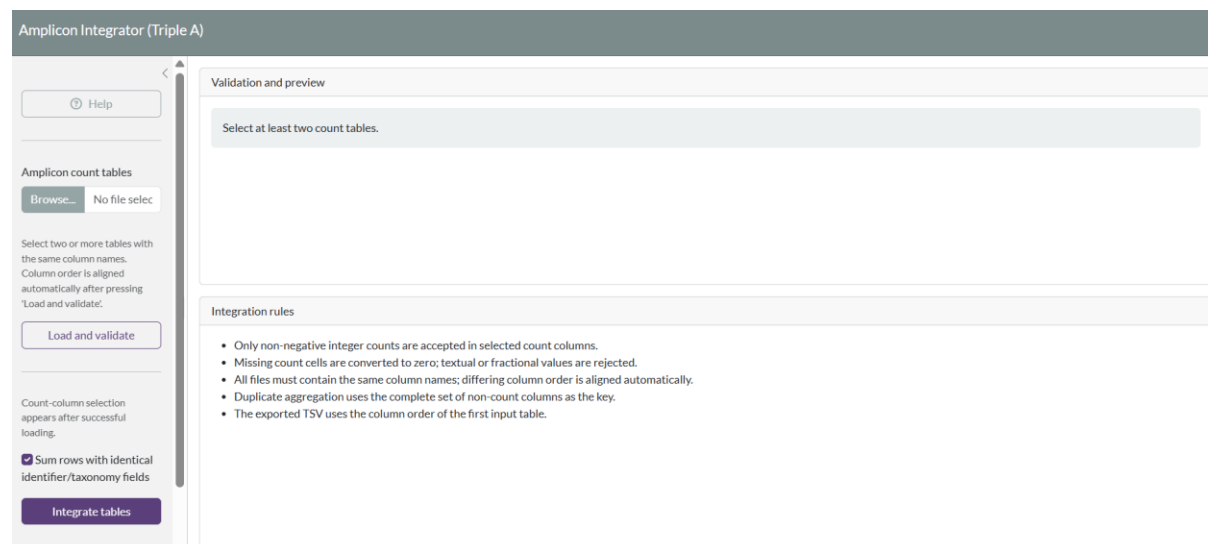


Figure 6. Amplicon Integrator module interface of Triple A

4.3. Reference Annotation Cache module

The Cache Module is responsible for improving the computational efficiency of Triple A by locally storing reference genome annotations and GFF (General Feature Format) files associated with previously analysed taxa. Functional profiling relies on identifying representative reference genomes and retrieving their genomic annotations from the NCBI databases, a process that may require hundreds of individual queries when analysing large microbial communities. By caching these annotation files after their first retrieval, Triple A eliminates redundant NCBI requests in subsequent analyses involving the same taxa. This strategy substantially reduces execution time, decreases network dependency, and enables rapid and fully reproducible functional analyses without modifying the underlying analytical workflow. The cache operates transparently to the user and is managed automatically.

The module performs:

- Data importation and merging
- Data exportation
- Cache integrity validation
- Cache curation

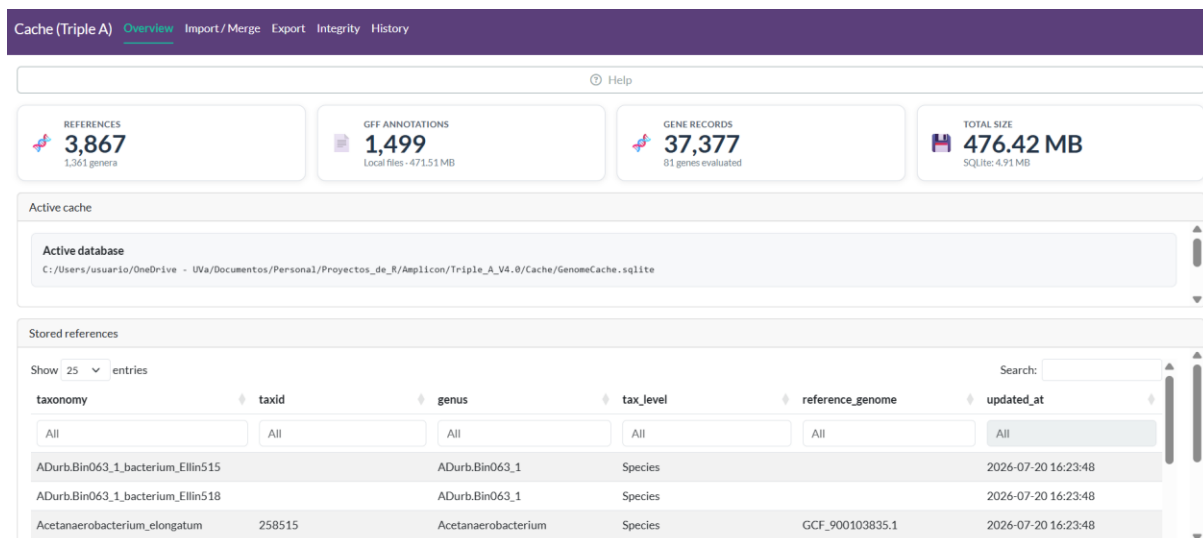


Figure 7. Cache module interface of Triple A

4.4. Biological Function Builder module

This module allows the user to add new biological functions and pathways to the curated registry declaratively, without editing R code. A function is described by its diagnostic, supporting and accessory genes and by the rule that combines them; the definition is stored as JSON under Resources/FunctionalDB/CustomFunctions/ and validated before it is accepted.

Custom functions appear alongside the curated ones in the functional-potential analysis. Gene synonyms can be declared too, so a marker named differently in an annotation is still found.

Figure 8. Biological Function Builder module interface of Triple A

4.5. Biological Analysis module

The main application, and where most of a study happens. It follows a fixed order, and each step unlocks the next:

- Import the abundance table and, optionally, the sample metadata.
- Declare the abundance scale (counts, percentage or proportion). The declaration is checked against the data rather than trusted.

- Assign column roles: sample identifier, grouping column, experimental factor, environmental variable.
- Declare the experimental design, from a metadata column or by column order.
- Read the design summary, which states in plain text how many groups were found and how many samples each has.
- Select analyses and figures. Unavailable analyses stay visibly locked and explain how to unlock themselves.
- Run, then browse the results in the result tree.

Two controls are worth knowing. Progress details choose how much the run reports: 'standard' gives concise messages, 'detailed' reports one message per taxon or genome, and 'developer' includes every gene search, useful when a functional run behaves unexpectedly. Analysis history lets you reopen a previous run and reload its results without recomputing them.

Results are shown one at a time in a result tree rather than all at once, which keeps a large run navigable and the session responsive.

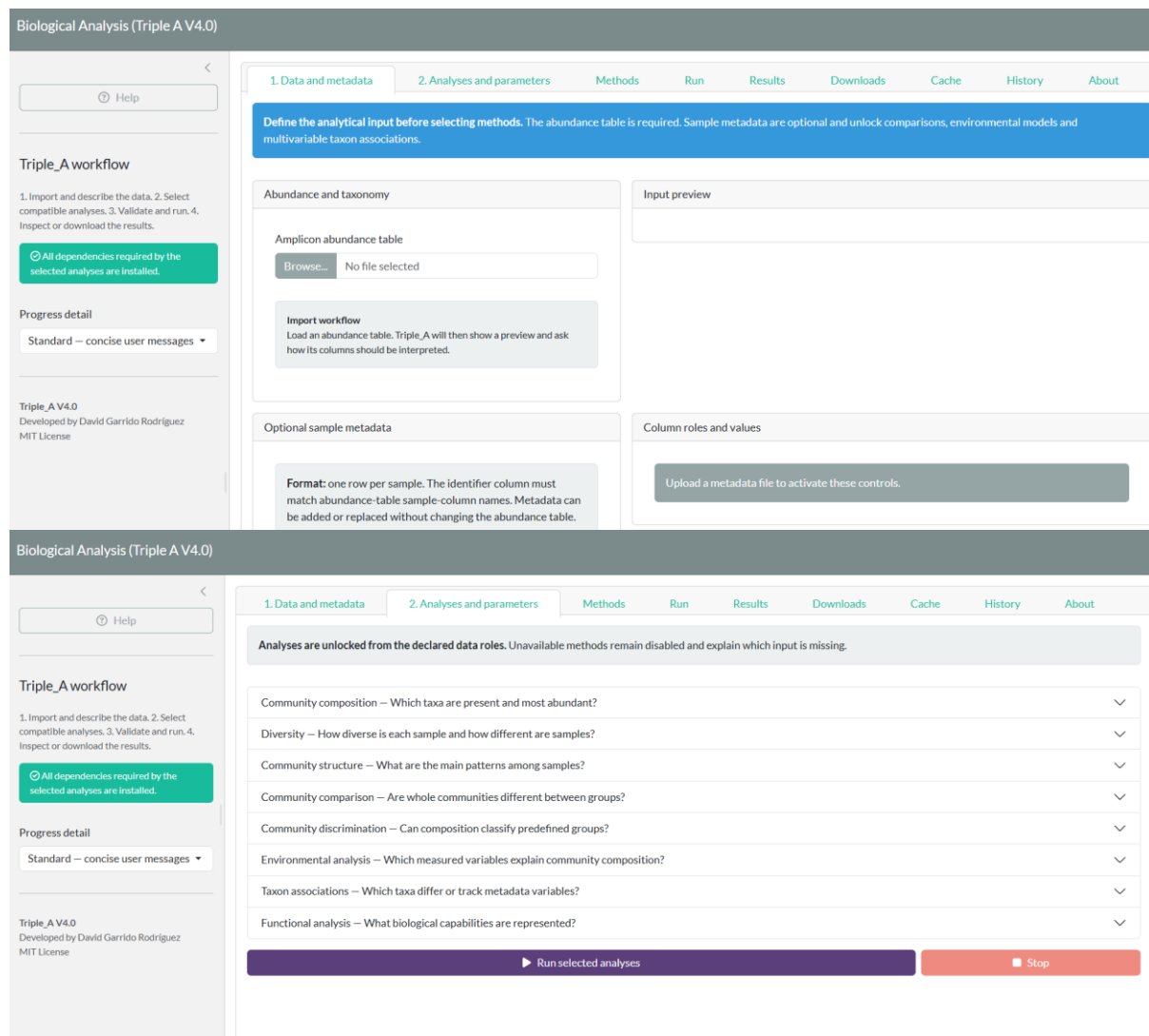


Figure 9. Biological Analysis module interface of Triple A

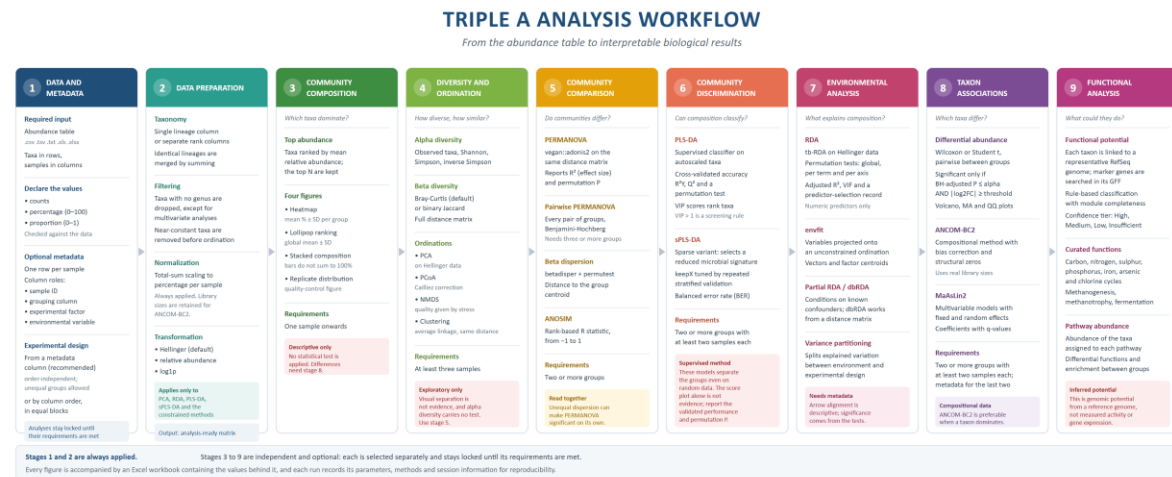


Figure 10. Detailed biological analysis workflow with all the possible analysis performed with Triple A

5. How do I use Triple A?

5.1. Input data

5.1.1. Abundance table

One row per taxon and one numeric column per sample. Accepted formats are .csv, .tsv, .txt, .xls and .xlsx. Taxonomy may be supplied either as a single lineage column or as separate rank columns. The abundance scale must be declared: counts, percentage or proportion. The declaration is checked against the data, and a table declared as counts but containing non-integer values is rejected rather than silently coerced. This matters because ANCOM-BC2 models sequencing depth and must receive real counts.

Table 4. Input abundance table reference

Taxonomy	Sample_A1	Sample_A2	Sample_A3	Sample_A4	Sample_B1	...
k_Bacteria;p_Phylum1;c_Class1;o_Order1;f_Family1;g_Genus1	19	19	19	23	49	...
k_Bacteria;p_Phylum2;c_Class2;o_Order2;f_Family2;g_Genus2	23	20	16	14	51	...
k_Bacteria;p_Phylum3;c_Class3;o_Order3;f_Family3;g_Genus3	24	18	21	21	53	...
k_Bacteria;p_Phylum4;c_Class4;o_Order4;f_Family4;g_Genus4	21	22	19	20	47	...

Table 5. Abundance type declaration

Abundance type	Expected input	Processing performed by Triple A
counts	Integer read counts for each sample	Each sample is normalized to relative abundance using Total Sum Scaling (TSS) , resulting in percentages.
percentage	Values ranging from 0 to 100, with each sample summing to approximately 100	Values are used directly without further scaling.
proportion	Values ranging from 0 to 1, with each sample summing to approximately 1	Values are multiplied by 100 to obtain percentages.

Practical Implications: Regardless of the input format, all abundance data are internally converted to relative abundances expressed as percentages on a per-sample basis. Consequently, **every downstream analysis performed by Triple A is based on community composition rather than absolute microbial abundance.**

This normalization strategy is the standard approach for amplicon sequencing data and is appropriate for compositional analyses. However, it also means that Triple A cannot be used to infer absolute abundance changes. For example, if the total microbial biomass of a community doubles while its taxonomic composition remains unchanged, the resulting abundance of profiles, and therefore all graphical outputs and statistical analyses, will be identical.

Validation rules:

- Taxonomic information may be provided either as a single semicolon-separated lineage (as shown above) or as separate taxonomic rank columns (e.g., Kingdom, Phylum, Class, etc.). In the latter case, the import wizard prompts the user to confirm the column mapping.

* Taxonomic data with confidence value

Triple A does not currently support taxonomic annotations that include confidence values within the taxonomy string, for example:

`d__Bacteria(100);p__Phylum(100);c__Class(100);o__Order(100);f__Family(100);g__Genus(100)`,

Before importing the dataset, these confidence values must be removed so that the taxonomy follows the standard lineage format expected by Triple A, for example:

`d__Bacteria;p__Phylum;c__Class;o__Order;f__Family;g__Genus`

The confidence values can be removed automatically using the following R code:

```
library(readxl)
df <- readxl::read_xlsx("file_path")
df$Taxonomy <- gsub("\\([\\^)]*\\)", "", df$Taxonomy)
write.csv(df, "new_file_path", row.names = FALSE)
```

This script removes any confidence values enclosed in parentheses while preserving the taxonomic hierarchy, producing an abundance table that is fully compatible with Triple A.

- Sample identifiers must be unique and non-empty.
- All sample columns must contain non-negative numerical values.
- Rows sharing an identical taxonomic lineage are automatically merged by summing their abundances; therefore, prior deduplication is not required.
- For all analyses except multivariate analyses, taxa without an assigned genus (i.e., an empty Genus field or a value of "None") are removed during preprocessing. Multivariate analyses, including PCA, PCoA, NMDS, RDA, PLS-DA, and sPLS-DA, are performed using the complete abundance table.

5.1.2. Experimental design

The design can be declared in two ways, which are equivalent and produce identical results:

Metadata column: *group membership is matched by sample identifier. Abundance columns may be in any order, groups may be interleaved, and groups may have different numbers of replicates.*

Block mode: *consecutive blocks of columns form each group. This is the default and remains unchanged.*

Group order sets the reference level of every pairwise comparison.

Caution when using block mode

If three treatments with four replicates are declared and named "Control, Treated, Recovery", the first four selected columns become Control, the next four Treated, and the last four Recovery, regardless of how those columns are named.

A table with interleaved samples (C1, T1, C2, T2, and so on) will produce a complete analysis, with figures of entirely normal appearance, and every treatment assigned incorrectly.

The application does block the case in which the number of columns does not match treatments x replicates, but it cannot detect an incorrect order with the correct number of columns. Option A should be used in that situation.

This mode additionally requires all groups to contain the same number of replicates (between one and five). Option A carries no such restriction.

5.1.3. Environmental metadata

A separate file with one row per sample. It requires an identifier column whose values match the sample column names of the abundance table. Matching tolerates differences in capitalization and separators, so that S-1, S.1, S_1 and s1 are treated as the same identifier.

Table 6. Example: *environmental_metadata.csv*

Sample	Treatment	Temperature	pH	Moisture	NH4	...
Sample A1	A	20.2	7.0	61	5.1	...
Sample A2	A	20.4	7.0	62	5.2	...
Sample A3	A	20.6	7.0	63	5.3	...
Sample B1	B	27.9	6.4	48	9.4	...

During import, each column is assigned a role:

- Identifier → exactly one column, the one matched against the samples.
- Environmental_variable → continuous variables. These are the predictors RDA can use.
- experimental_factor → categorical variables. Available to ANCOM-BC2, MaAsLin2, envfit and variance partitioning.
- Any column may additionally be used as the source of the experimental design, irrespective of the role assigned to it.

RDA requires strictly numeric variables: any column that cannot be converted to a number is rejected with an explicit error. A categorical variable coded as 1/2/3 will be accepted and treated as continuous, which is almost never the intended behaviour. For categorical predictors, use envfit, partial RDA or variance partitioning instead

5.1.4. Environmental-model defaults

Shared by RDA, partial RDA, dbRDA, partial dbRDA, envfit and variance partitioning.

Table 7. *Environmental model defaults*

Parameter	Default	Description
rda_permutations	999	Number of permutations for the significance tests of every environmental model. 999 gives a minimum attainable P of 0.001; raise it to 9999 for results near the threshold.
rda_alpha	0.05	Significance threshold applied to those permutation tests.

5.1.5. Minimum requirements and locked analyses

Table 8. Minimum requirements and locked analyses

Analysis	Minimum_requirement
Top abundance	At least one sample column.
Community structure	At least three samples. PERMANOVA, ANOSIM and beta dispersion need at least two groups; pairwise PERMANOVA needs at least three.
Differential abundance	At least two treatment groups with at least two replicates each.
PLS-DA / sPLS-DA	At least two groups with at least two replicates each, and at least three samples.
RDA	Environmental metadata with at least one numeric variable, matched sample identifiers and residual degrees of freedom.
envfit	Environmental metadata with at least one informative variable and at least four matched samples.
Variance partitioning	At least one environmental variable and at least one experimental factor.
ANCOM-BC2 / MaAsLin2	Sample metadata declaring at least one factor or environmental predictor.
Functional potential	An imported abundance table and at least one selected biological function.

An analysis whose requirements are not met is disabled and explains, under 'How to unlock', the steps that enable it. The same rules are re-checked when the run is launched, so an ineligible analysis cannot start. A sample missing from the metadata blocks the design rather than silently dropped.

5.2. Output files and figure location

Each figure generated by **Triple A** is accompanied by a standardized interpretation guide comprising: the output filename, what the figure represents, how it is generated, how it should be interpreted, what conclusions cannot be drawn from it, and the warning signs that may indicate an unreliable or misleading result.

The Interpretation and Limitations sections are particularly important. Most errors in amplicon data interpretation do not arise from misreading a figure, but from drawing conclusions that the analysis itself cannot support. Common examples include interpreting visual separation as evidence of statistical significance, equating genomic potential with biological activity, or confusing relative abundance with absolute abundance.

Each analysis run creates a dedicated directory under Results/Runs/, containing one subdirectory for each analysis module. Every module includes the corresponding graphical outputs (PNG format) together with an Excel workbook (.xlsx) containing all numerical data used to generate those figures.

Every figure is therefore fully supported by an underlying spreadsheet. Whenever the visual interpretation of a figure is uncertain, the numerical values stored in the accompanying

workbook, not the graphical representation itself, should be regarded as the definitive source of information.

Each run also writes the following files at the root of its own directory:

- Run_metadata.json
- Session_information.txt
- Analysis_methods.xlsx
- Triple_A_manifest.xlsx

The Analysis_methods.xlsx workbook is generated automatically and documents, for every selected analysis, its objective, data preprocessing workflow, statistical methodology, significance criteria, and known limitations. Together, these records provide a comprehensive description of the analytical workflow and can serve as a valuable starting point for drafting the Materials and Methods section of a scientific manuscript.

A self-contained Triple_A_report.html is written at the end of every run, gathering the parameters, the methodological summary and every figure embedded in the file itself. A PDF version is produced automatically when a Chrome installation is available; if it is not, the HTML can be printed to PDF from the browser.

5.3. How to read an ordination

Six of the figures produced by Triple A are scatter plots of samples on two axes: PCA, PCoA, NMDS, PLS-DA, sPLS-DA and RDA. They share the same plotting engine and the same conventions. The scale you are looking at is set by the ordination transformation (section 8.1).

5.3.1. Common conventions:

- Each point is a sample: color indicates the treatment; the symbol is always the same, because mapping shape as well would exhaust the discrete symbol palette when many groups are present.
- Ellipses are 95% confidence ellipses assuming a bivariate normal distribution. They are drawn only for groups with at least three distinct positions and a non-degenerate covariance matrix. A group without an ellipse is not a failure: it means there is not enough dispersion to estimate one.
- The grey horizontal and vertical lines mark the origin of coordinates.
- The subtitle reports how many samples are being displayed out of the total. If that number is lower than expected, some samples have missing coordinates and the reason should be investigated.

Points separated artificially

When two or more samples fall on exactly the same coordinate, Triple A separates them visually by displacing them in a circle by 1.4% of the axis range, and states this in the subtitle ("... samples in overlapping positions were separated visually").

That displacement is purely graphical: the coordinates exported to the workbook are the analytical ones. The small distance between those points must not be interpreted as biological variation.

5.3.2. The most common interpretation error

Visual separation between groups in an ordination is NOT statistical evidence. PCA, PCoA and NMDS are unsupervised, descriptive methods: they have no knowledge of the treatments, and colour is applied after the coordinates have been computed. Even so, with few samples and many taxa it is easy to obtain apparently separated clouds purely by chance.

Claiming that communities differ requires the corresponding test: PERMANOVA together with its beta-dispersion diagnostic, ANOSIM, or the permutation tests of the RDA. For PLS-DA and sPLS-DA the situation is more delicate still, because they are supervised methods: they do know the labels and will separate the groups even on entirely random data. There, the only informative quantity is cross-validated performance.

5.3.3. Axis percentages

- In PCA and PCoA the axis percentage is the fraction of the total variation that the axis represents. Axes with low percentages, below roughly 10%, describe very little: two samples far apart on an axis explaining 5% are barely different.
- In NMDS the axes carry no associated percentage and are not individually interpretable: only the relative distances between points matter, and overall quality is given by the stress value.
- In RDA there are two kinds of axis and the figure distinguishes them in the label. RDA axes report "% constrained", that is, the percentage of the variation EXPLAINED BY THE ENVIRONMENTAL VARIABLES, not of the total. PC axes, where present, report "% total; unconstrained". The distinction matters: an RDA axis capturing 85% of the constrained variation may represent only 19% of the total.

5.3.4. Reading results well

- Report effect sizes alongside P-values, and adjusted P-values rather than raw ones.
- Read PERMANOVA together with its beta-dispersion diagnostic: a significant PERMANOVA with significant dispersion may reflect a difference in spread, not in centroid.
- Functional results describe inferred genomic potential from a reference genome, never measured activity or gene expression.
- Every figure has a workbook behind it. When a visual reading is uncertain, the numbers in the workbook are the result, not the picture

6. Community composition

Biological question

Which microorganisms compose the community, in what proportions, and which taxa dominate the samples?

A purely descriptive module. Its purpose is to establish what the community is made of before any inference is attempted. Read it as a description of relative composition, a taxon can appear dominant simply because sequencing depth or primer bias favoured it. It ranks taxa by mean abundance and displays the top N. None of these figures contains statistical inference. Beyond about 40 the heatmap labels become unreadable.

Table 9. Top-taxa heatmap plot

Output file	Top_taxa_heatmap.png
What it shows	A taxon x treatment matrix with the mean relative abundance of each taxon in each treatment. Each cell is annotated with "mean % $\pm$ standard deviation %" across replicates.
How it is computed	Mean and standard deviation of the replicates of each treatment on relative abundance (%). No hierarchical clustering: rows follow the global mean-abundance ranking and columns the declared treatment order.
How to read it	Colour intensity encodes the magnitude of abundance. The informative comparison is within a row, that is, the same taxon across treatments. The printed standard deviation indicates replicate consistency: a high mean with a deviation of comparable magnitude means a single replicate is driving the value.
What it does not prove	It does not demonstrate differences between treatments; the differential-abundance module is required for that. Nor is it absolute abundance.
Warning signs	Standard deviations comparable to or larger than the means across several rows indicate poorly homogeneous replicates, which compromises the entire downstream analysis, not only this figure.

Table 10. Top-taxa lollipop plot

Output file	Top_taxa_lollipop.png
What it shows	A horizontal ranking of the selected top N taxa by global mean abundance, with error bars of one standard deviation.
How it is computed	Mean and standard deviation of the relative abundance of each taxon across ALL samples, without separating by treatment. Error bars are clipped at zero.
How to read it	It gives the dominance hierarchy of the experiment as a whole: which taxa structure the community. Error-bar length measures variability across all samples, which includes both between-replicate and between-treatment variation.
What it does not prove	It does not separate by treatment: a taxon with a very long error bar may be strongly differential between groups, and this figure cannot distinguish that from a merely noisy taxon.
Warning signs	If the first two or three taxa account for almost all abundance, the community is dominated by a few organisms and the ordinations will be governed by them. This should be checked against the PCA contribution plot.

Table 11. Top-taxa stacked composition plot

Output file	Top_taxa_stacked_composition.png
What it shows	One stacked bar per treatment; each segment is a taxon from the top N.
How it is computed	Mean relative abundance (%) across replicates, by treatment and taxon.
How to read it	It shows the compositional distribution and how it rearranges between treatments. A change in segment height between bars is a change in relative abundance.
What it does not prove	The bars do NOT sum to 100%: only the selected top N taxa are represented and the remainder of the community is omitted, as stated in the subtitle. It must not be read as a complete composition.

Warning signs	If the bars sum to markedly different values across treatments, the top N captures a very different fraction of each community, which indicates that N is too small for the intended comparison.
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Table 12. Top-taxa distribution plot

Output file	Top_taxa_distribution.png
What it shows	A box plot per taxon with the individual sample points overlaid, coloured by treatment.
How it is computed	Sample-level relative abundance values, without averaging. Points are jittered horizontally to avoid overlap.
How to read it	This is the quality-control figure of the module: it exposes the real dispersion behind the means shown in the other three. Look for replicates that separate clearly from their group companions, which are candidates for outliers or labelling errors.
What it does not prove	The box carries no associated test. Non-overlapping boxes do not imply significance.
Warning signs	A point from one treatment consistently positioned among those of another group, across several taxa at once, suggests a swapped sample. Verify the experimental design.

Table 13. Top-abundance quick reference

Community composition – Top abundance	
When to use it	As the first descriptive step, before any test, to see what the community is made of.
Why Triple A implements it this way	It ranks taxa by mean relative abundance and shows the top N, because dominance and statistical difference are separate questions.
Typical interpretation (example)	The community was dominated by Genus X (mean 24% +/- 3), followed by Genus Y (11% +/- 2).
Common mistakes	Reading dominance as a treatment difference; forgetting the bars omit the rest of the community.
Complementary analyses	Differential abundance (for differences); PCA contribution plot (whether dominants drive the ordination).
Reporting recommendations	Report the top taxa as mean +/- SD per group, the N shown, and state it is relative abundance.
Take-home message	It describes what the community is made of, not whether groups differ.

7. Diversity

Biological question

How many taxa occur within samples, how evenly distributed are they, and how dissimilar are communities between samples?

7.1. Alpha diversity

Summarises diversity within each sample: observed richness (how many taxa), Shannon and Simpson (how evenly abundance is spread) and inverse Simpson. Richness is the most sensitive to sequencing depth, because deeper sequencing finds more rare taxa; the evenness-weighted indices are more robust. Report which index you used, since they can rank samples differently when a community is dominated by a few taxa. Output files are a graph and a table.

Table 14. *alpha diversity plot*

Output file	Alpha_diversity.png
What it shows	Four panels (observed taxa, Shannon, Simpson and inverse Simpson) with boxes by treatment and the individual sample points.
How it is computed	vegan::diversity on relative abundance. Observed taxa is the count of taxa with abundance greater than zero.
How to read it	Shannon combines richness and evenness; Simpson weights towards dominant taxa; inverse Simpson is interpreted as an effective number of taxa. When Shannon rises but richness does not, what has increased is evenness.
What it does not prove	This figure carries NO associated test, as explicitly declared in the application's methodology registry. Non-overlapping boxes do not license the claim that diversity differs. A separate test, such as ANOVA or Kruskal-Wallis, run outside Triple A, is required.
Warning signs	The observed-taxa count is neither rarefied nor corrected for sequencing depth. If samples were sequenced at different depths, richness differences may be purely technical. Shannon and Simpson are considerably more robust in this respect.

Table 15. *alpha diversity table*

Output file	Community_structure_summary.xlsx (Alpha_diversity sheet).
What it contains	One row per sample with Observed taxa, Shannon, Simpson and Inverse Simpson, plus Treatment and Replicate.
How it is computed	vegan::diversity on relative abundance. Observed_taxa counts taxa with abundance > 0.
How to read it	Compare indices across groups; report which index you used.
What it does not prove	No test is included; observed richness is depth-sensitive. Test with ANOVA/Kruskal-Wallis outside Triple A.

Table 16. *Alpha diversity quick reference*

Alpha diversity	
Biological question	How rich and even is each sample internally?
When to use it	To compare within-sample diversity across groups or gradients.
Why Triple A implements it this way	It reports observed richness, Shannon, Simpson and inverse Simpson on relative abundance, and imposes no test because the right test depends on the design.
Typical interpretation (example)	Shannon diversity was higher in treated than in control samples (mean 3.6 vs 3.1).
Common mistakes	Claiming a difference from non-overlapping boxes without a test; comparing richness across unevenly sequenced samples.
Complementary analyses	A separate ANOVA or Kruskal-Wallis (run outside Triple A); a sequencing-depth check for richness.
Reporting recommendations	State the index used and its value per group, plus the external test statistic and P.
Take-home message	Diversity indices describe samples; significance needs a test you run yourself.

7.2. Beta diversity

Quantifies how dissimilar samples are from each other, producing the distance matrix that every ordination and composition test in the following panels is built on. Bray-Curtis uses abundances and is the default, while Jaccard uses presence and absence only and is the right choice when abundance estimates are unreliable. **The choice made here propagates through PCoA, NMDS, PERMANOVA, ANOSIM and dbRDA, so it should be made deliberately.** Output file is a table (smaller values indicate more similar samples under the selected metric).

Table 17. *beta distance matrix table*

Output file	Community_structure_summary.xlsx (Beta_distance sheet).
What it contains	Symmetric sample-by-sample dissimilarity matrix; the substrate of every ordination and composition test.
How it is computed	vegan::vegdist on relative abundance (Bray-Curtis default, or binary Jaccard).
How to read it	Smaller = more similar; diagonal is zero.
What it does not prove	The metric choice propagates downstream; Jaccard over-weights rare taxa.

Table 18. *beta distance matrix quick reference*

Beta diversity	
Biological question	How dissimilar are samples from each other?
When to use it	Whenever a community-level comparison or ordination is needed; it underlies everything downstream.
Why Triple A implements it this way	vegan::vegdist with Bray-Curtis (default) or binary Jaccard, chosen once so the same dissimilarity propagates consistently.
Typical interpretation (example)	Bray-Curtis dissimilarities were larger between than within treatments.
Common mistakes	Switching metric without noting it; ignoring that Jaccard over-weights rare taxa.
Complementary analyses	PCoA, NMDS, PERMANOVA and dbRDA all read this matrix.
Reporting recommendations	Name the dissimilarity metric and the abundance scale used.
Take-home message	The distance matrix is the foundation of every ordination and composition test.

8. Community structure

Biological question

Do samples form gradients, clusters or recurring compositional patterns without imposing an environmental model?

8.1. Ordination transformation

Why are the abundances transformed before an ordination?

Amplicon abundance data are compositional and strongly skewed: a handful of taxa usually account for most of the reads. Left untransformed, the distances between samples are driven almost entirely by those few dominant taxa, and the ordination shows their variation rather than the community's. Transforming rescales the table so that moderately abundant taxa also contribute, without discarding the abundance information altogether. The transformation is applied before the distance matrix is computed, so it shapes not only the ordinations but also the composition tests built on those distances.

Table 19. *Transformation methods*

Transformation	Meaning
hellinger	Square root of relative abundance. Default. Reduces the weight of dominant taxa while keeping Euclidean distances meaningful.
relative	Proportion of the sample total. No variance stabilisation.
log1p	$\log(1 + x)$. Compresses large values; useful for strongly skewed count data.

* Hellinger (the square root of relative abundance) is the default and the recommended choice: it makes the Euclidean distance implicit in PCA and RDA equivalent to the Hellinger distance, which is appropriate for compositional data. This is standard practice (Legendre and Gallagher, 2001) and there is no reason to change it unless the rationale for doing so is well understood.

Applies to ordination methods based on transformed abundance matrices (PCA, PLS-DA, sPLS-DA, RDA and constrained ordinations), but not to distance-based methods such as PCoA or NMDS

8.2. Principal component analysis (PCA)

A linear ordination of the transformed abundance matrix that places samples so the first axes capture the largest gradients of variation. It is unconstrained and descriptive, nothing about the experimental design enters the computation, which is precisely what makes it useful as an unbiased first look. The accompanying scree plot shows how much variation each axis carries.

Table 20. PCA sample plot

Output file	PCA.png
What it shows	Samples projected onto the first two principal components. The axes report the percentage of variance they represent.
How it is computed	precomp (centred, unscaled) on the transformed matrix. With the default Hellinger transformation this is exactly a Hellinger PCA, which is appropriate for compositional data. Taxa are not scaled to unit variance: abundant taxa weigh more than rare ones, which is the desirable behaviour here.
How to read it	Proximity means similar transformed profiles. The axis percentages bound how much of the real variation is visible: if PC1 and PC2 sum to 30%, the figure hides 70% of the structure.
What it does not prove	It is not a test. See the chapter on reading ordinations.
Warning signs	PC1 explaining more than 80% usually means a single dominant taxon governs the ordination; check this against the contribution plot. If one point lies far from the rest, PC1 is describing that outlier rather than the experimental effect.

Table 21. PCA scree plot

Output file	PCA_scree.png
What it shows	Percentage of variance explained by each of the first ten components, as bars with a superimposed line.
How it is computed	$100 \times (\text{component standard deviation})^2 / \text{sum of all}$.
How to read it	It informs how many components are worth examining. The usual criterion is the elbow: the point beyond which the curve flattens. Components to the right of the elbow describe noise.
What it does not prove	There is no universal threshold for an acceptable percentage; it depends on the number of taxa and samples.
Warning signs	A curve that is nearly flat from the first component means there is no dominant structure in the data: PCA will not summarize anything useful and PCoA or NMDS with an ecological distance should be used instead.

Table 22. PCA taxon contributions

Output file	PCA_loadings.png
What it shows	The twenty taxa with the largest joint contribution to PC1 and PC2, as horizontal bars.
How it is computed	The plotted metric is the sum of the squared loadings on PC1 and PC2. The workbook additionally reports the percentage contribution separately for each component.
How to read it	It identifies which taxa are generating the ordination. This is the obligatory step after observing a separation in the PCA: if the responsible taxa make no biological sense in the context of the experiment, the separation is probably technical.
What it does not prove	It does not indicate the direction in which each taxon pushes, because the metric is a square and therefore always positive. For direction, consult the sign of PC1 and PC2 in the PCA loadings sheet of the workbook.
Warning signs	This column is labelled Cos2 in the workbook. Strictly it is not a squared cosine but a loading-based contribution; the ranking is valid but the name should not be cited as cos2 in a manuscript.

Table 23. PCA quick reference

PCA	
Biological question	Do samples naturally cluster without considering experimental groups?
When to use it	As an unbiased first look at the main gradients of variation.
Why Triple A implements it this way	prcomp (centred, unscaled) on the Hellinger-transformed table, so the implicit Euclidean distance equals the Hellinger distance appropriate for compositional data (Legendre & Gallagher 2001).
Typical interpretation (example)	PC1 (38%) separated saline from freshwater samples, indicating that salinity was the main source of compositional variation.
Common mistakes	Reading visual separation as significance; interpreting axes that explain little (< 10%).
Complementary analyses	PCA contribution/loadings plot (which taxa drive it); PERMANOVA (for the test).
Reporting recommendations	Report the % variance of PC1 and PC2, the transformation, and that it is descriptive.
Take-home message	PCA shows structure, never significance.

8.3. Distance-based ordinations

8.3.1. Principal coordinates analysis (PCoA)

The same idea as PCA but applied to the beta-diversity distance matrix instead of the raw variables. This matters because it lets the ordination honour an ecological dissimilarity such as Bray-Curtis, which PCA cannot represent directly.

Table 24. PCoA plot

Output file	PCoA.png
What it shows	A two-dimensional representation of the matrix of ecological distances between samples. The title states the metric used.
How it is computed	vegdist with Bray-Curtis (default) or binary Jaccard on relative abundance, followed by metric multidimensional scaling with the Cailliez correction (add=TRUE) to remove negative eigenvalues. Percentages are computed over the positive eigenvalues.
How to read it	As for PCA, but respecting an ecologically appropriate distance rather than the Euclidean one. Bray-Curtis weights abundances; Jaccard considers only presence and absence and therefore gives far more weight to rare taxa.
What it does not prove	It is not a test. It is also the sibling figure of PERMANOVA: both use exactly the same distance matrix, so they must be read together.
Warning signs	If PCoA and NMDS give contradictory pictures, trust the NMDS when its stress is low: it means the distance structure is not well represented metrically.

Table 25. PCoA quick reference

PCoA	
Biological question	Do samples cluster under an ecological distance rather than a Euclidean one?
When to use it	When Bray-Curtis or Jaccard is more appropriate than the Euclidean geometry PCA assumes.
Why Triple A implements it this way	cmdscale with the Cailliez correction (add=TRUE) on the beta-distance matrix, so it honours the chosen dissimilarity.
Typical interpretation (example)	PCoA on Bray-Curtis distances showed treatment groups occupying distinct regions of the ordination.
Common mistakes	Treating it as a test; forgetting it and PERMANOVA share the same distance matrix.
Complementary analyses	PERMANOVA (same matrix, provides the test); NMDS (rank-based cross-check).
Reporting recommendations	Report the distance metric and the % of positive eigenvalues on each axis.

PCoA	
Take-home message	PCoA and PERMANOVA are two views of one distance matrix; read them together.

8.3.2. Non-metric multidimensional scaling (NMDS)

Arranges samples so the rank order of the distances between them is preserved as well as possible in two dimensions. Because it uses ranks rather than absolute values, it copes well with ecological data. Its axes carry no percentage of variation; the diagnostic to read instead is the stress value, where below 0.2 is normally considered a usable representation.

Table 26. NMDS plot

Output file	NMDS.png
What it shows	An ordination that preserves the RANK ORDER of distances between samples rather than their magnitude. The title includes the stress value.
How it is computed	metaMDS with k=2, without automatic transformation because the data are already normalized, using the number of starts configured in nmms_rymax. The seed is fixed, so the figure is reproducible.
How to read it	Stress is the quality indicator and appears in the title. Conventional guidance: below 0.10 is a good representation; 0.10 to 0.20 is usable with caution; above 0.20 is a poor representation and positions should not be interpreted in detail. In NMDS the axes have no individual meaning and the orientation is arbitrary.
What it does not prove	The axes carry no percentage and no interpretation. It is incorrect to state that "NMDS1 separates the treatments".
Warning signs	A stress of exactly zero, or very close to it, with few samples is a degenerate solution rather than an excellent result. The nmms_diagnostics sheet of the workbook records any convergence warning issued by metaMDS and should always be consulted.

Table 27. NMDS quick reference

NMDS	
Biological question	Does the rank order of dissimilarities reveal grouping in two dimensions?
When to use it	When a robust, rank-based ordination is preferred for noisy ecological data.
Why Triple A implements it this way	metaMDS with k=2, autotransform off (data already normalized) and a fixed seed for reproducibility; quality is read from stress, not axes.
Typical interpretation (example)	NMDS (stress = 0.08) separated the communities by treatment.
Common mistakes	Saying 'NMDS1 separates treatments' (axes have no meaning); trusting a near-zero stress with few samples (degenerate).
Complementary analyses	PERMANOVA (the test); the nmms_diagnostics sheet (convergence).
Reporting recommendations	Report the stress value; never cite axis percentages (there are none).
Take-home message	Only stress and relative distances matter; the axes do not.

8.3.3. Hierarchical clustering

Groups samples by successively joining the least dissimilar ones, using the same distance matrix as the ordinations, so the dendrogram and the PCoA cannot disagree about how far apart samples are. Linkage is average (UPGMA), the conventional choice for ecological dissimilarities because it does not assume Euclidean geometry the way Ward's method does. It answers a different question from an ordination: not 'how do samples spread out' but 'which samples join first'. Read the height at which two branches merge, not the left-to-right order of the leaves, which is arbitrary among equivalent rotations of the tree.

The accompanying table cuts the dendrogram at the number of declared treatment groups, so you can compare cluster membership against the design directly. Treat all of this as exploratory: clustering always produces groups, whether or not they are real, and agreement with the design is suggestive rather than a test, use PERMANOVA for that.

Table 28. Hierarchical clustering plot

Output file	Hierarchical_clustering.png
What it shows	A tree in which samples are joined successively by community dissimilarity. The height at which two branches merge is the dissimilarity at which those samples join: low merges mean similar communities. Leaves are coloured by declared treatment, and the companion table gives each sample's cluster.
How it is computed	stats::hclust with average linkage (UPGMA) on the same Bray-Curtis (or Jaccard) distance matrix used by PCoA, NMDS and PERMANOVA, so the dendrogram cannot disagree with them. That matrix is built from relative abundances, which means the <i>Ordination transformation</i> setting does not change this figure. The companion table cuts the tree with stats::cutree at k = the number of declared treatment groups.
How to read it	Read the height at which branches merge, never the left-to-right order of the leaves: any branch can be rotated about its node and still be the same tree. Compare the branch structure against the leaf colours to see whether an unsupervised grouping recovers the declared design.
What it does not prove	It is not a test and yields no P-value. Clustering always produces clusters, including random data. Agreement between branches and treatments is suggestive, not evidence, PERMANOVA is the test for that. The number of clusters in the table is imposed (k = declared groups), not discovered: it does not mean the data support that many.
Warning signs	Merge heights that are all very small: samples are nearly identical and the structure shown may be noise. One sample joining last at a much greater height: a likely outlier, check its sequencing depth. A tree that changes substantially when the distance is switched between Bray-Curtis and Jaccard: the structure is not robust. Branches that cut across treatments while PERMANOVA is significant: read the beta-dispersion diagnostic, as the difference may be in spread rather than in composition. Average linkage is one choice among several; a different linkage can yield a different tree from identical distances.

Table 29. Hierarchical clustering table

Output file	Community_structure_summary.xlsx (Hierarchical_clustering sheet).
What it contains	Each sample with its Treatment and assigned cluster.
How it is computed	stats::cutree on the UPGMA dendrogram at k = number of declared groups.
How to read it	Compare cluster membership against the design.
What it does not prove	k is imposed, not discovered; agreement is suggestive, not a test.

Table 30. Hierarchical clustering quick reference

Hierarchical clustering	
Biological question	Which samples join first by community dissimilarity?
When to use it	As an exploratory complement to ordination, to see nested similarity.
Why Triple A implements it this way	hclust with average linkage (UPGMA) on the same distance matrix as PCoA/PERMANOVA, the conventional choice for ecological dissimilarities.
Typical interpretation (example)	Samples clustered by treatment, with control replicates forming a single clade.
Common mistakes	Reading the left-to-right leaf order (arbitrary); treating agreement with the design as a test.
Complementary analyses	PERMANOVA (the test); PCoA (same distances, continuous view).
Reporting recommendations	Report the linkage and distance; describe merge heights, not leaf order.
Take-home message	Clustering always produces clusters; it suggests, it does not test.

9. Community comparison

Biological question

Does community composition differ between experimental groups, and are significant results compatible with similar within-group dispersion?

9.1. PERMANOVA (Permutational multivariate analysis of variance)

Tests on whether the group centroids differ in the distance matrix, by comparing the observed between-group separation against a null distribution built by shuffling group labels. It reports R-squared as effect size, which should always be reported alongside the P-value: with enough samples a trivially small difference can be significant. This is the test that turns a visual separation in an ordination into evidence.

PERMANOVA answers the question the ordination cannot: Do the communities actually differ between treatments? It is computed on the same distance matrix as PCoA and NMDS.

Table 31. PERMANOVA plot

Output file	PERMANOVA_variance.png
What it shows	Horizontal bars with the R2 of each model term; the Total row is excluded.
How it is computed	R2 obtained from <code>adonis2(distance ~ Treatment)</code> with the configured number of permutations. The figure runs no additional test.
How to read it	R2 is the fraction of compositional variation attributable to the treatment. It is the effect size, and it is what should be reported alongside the P value. An R2 of 0.54 with P = 0.005 is a strong result; an R2 of 0.08 with P = 0.04 is significant but explains very little.
What it does not prove	The figure does not display the P value. It is in the PERMANOVA sheet of the workbook, together with a Significant column already evaluated against the configured alpha.
Warning signs	With few groups and few replicates R2 is mechanically biased upwards. Always interpret it together with the beta-dispersion result.

Table 32. PERMANOVA table

Output file	Community_structure_summary.xlsx (Beta_distance sheet).
What it contains	Symmetric sample-by-sample dissimilarity matrix; the substrate of every ordination and composition test.
How it is computed	<code>vegan::vegdist</code> on relative abundance (Bray-Curtis default, or binary Jaccard).
How to read it	Smaller = more similar; diagonal is zero.
What it does not prove	The metric choice propagates downstream; Jaccard over-weights rare taxa.

Table 33. PERMANOVA quick reference

PERMANOVA	
Biological question	Are the observed differences between treatments statistically significant?
When to use it	To formally test the group separation an ordination only suggests.
Why Triple A implements it this way	<code>adonis2</code> on the beta-distance matrix by permutation, reporting R-squared as effect size, the honest currency of multivariate difference.
Typical interpretation (example)	Community composition differed significantly between treatments (PERMANOVA, R2 = 0.54, P = 0.002).
Common mistakes	Reporting P without R-squared; ignoring the equal-dispersion assumption.
Complementary analyses	Beta dispersion (the assumption); pairwise PERMANOVA (which pairs).

PERMANOVA	
Reporting recommendations	Report R-squared, pseudo-F, P, the number of permutations and the distance metric.
Take-home message	The test that turns visual separation into evidence, but only alongside beta dispersion.

9.2. Analysis of similarities (ANOSIM)

A rank-based alternative that compares between-group and within-group dissimilarities and summarises the contrast as an R statistic between -1 and 1. It is more sensitive than PERMANOVA to differences in dispersion, so a disagreement between the two is informative rather than contradictory.

Table 34. ANOSIM plot

Output file	ANOSIM_R.png
What it shows	A gauge placing the ANOSIM R statistic on the -1 to 1 scale. The subtitle reports R, the P value and the number of permutations.
How it is computed	vegan::anosim on the same distance matrix. The marker colour changes with magnitude: R >= 0.75 strong separation, >= 0.25 moderate, below that weak.
How to read it	R compares between-group with within-group distances using ranks. R near zero means samples of the same group are no more similar to each other than to samples of other groups. R near one means complete separation.
What it does not prove	ANOSIM is sensitive to unequal group sizes and to differences in dispersion, more so than PERMANOVA. When the two disagree, PERMANOVA is preferable; ANOSIM is included because it remains common in the microbial ecology literature.
Warning signs	A high R with a non-significant P value is the expected pattern with very few replicates: there are not enough possible permutations to reach significance.

Table 35. ANOSIM quick reference

ANOSIM	
Biological question	Are between-group dissimilarities larger than within-group ones?
When to use it	When a rank-based contrast is wanted, or for comparability with older literature.
Why Triple A implements it this way	vegan::anosim on the same distance matrix; R from -1 to 1 summarises the contrast.
Typical interpretation (example)	ANOSIM confirmed group separation (R = 0.99, P = 0.001).
Common mistakes	Preferring ANOSIM over PERMANOVA when they disagree (it is more sensitive to dispersion and unequal group sizes).
Complementary analyses	PERMANOVA (preferred on disagreement); beta dispersion.
Reporting recommendations	Report R, P and the number of permutations.
Take-home message	A rank-based companion to PERMANOVA, not a replacement.

9.3. Beta dispersion

Tests on whether the groups differ in how spread out their samples are around their own centroid. This is the essential companion to PERMANOVA, not an optional extra: PERMANOVA is sensitive to dispersion, so a significant PERMANOVA accompanied by significant dispersion may reflect groups of different variability rather than groups of different composition. Always read the two together before concluding that composition differs.

Table 36. Beta dispersion plot

Output file	Beta_dispersion_boxplot.png
What it shows	Distance from each sample to its group centroid, by treatment. Boxes, individual points, and the mean marked with a diamond.
How it is computed	vegan::betadisper on the same distance matrix, in PCoA space. The associated permutation test is in the Beta_dispersion sheet of the workbook.
How to read it	This is the most important diagnostic figure of the module and the most frequently ignored. PERMANOVA assumes that dispersion within groups is homogeneous. If one group is far more dispersed than another, PERMANOVA may return a significant result because of that difference in dispersion rather than a shift in centroids.
What it does not prove	It does not replace PERMANOVA: it answers a different question, namely whether the groups are equally heterogeneous, as opposed to whether they are in different locations.
Warning signs	Markedly different box heights between groups, or a significant dispersion test, require the PERMANOVA to be qualified: the correct conclusion becomes that the communities differ in composition and/or in variability. This is also a biologically interesting result in itself: a perturbation that increases heterogeneity between replicates often indicates destabilization of the community.

Table 37. Beta dispersion quick reference

Beta dispersion	
Biological question	Do groups differ in how spread out their samples are?
When to use it	Always, whenever PERMANOVA or ANOSIM is run; it is the assumption check, not an optional extra.
Why Triple A implements it this way	vegan::betadisper in PCoA space, because PERMANOVA can be significant through unequal dispersion rather than different centroids.
Typical interpretation (example)	Dispersion did not differ between groups (betadisper, $P = 0.19$), so the PERMANOVA reflects a genuine compositional shift.
Common mistakes	Reporting a significant PERMANOVA without checking dispersion.
Complementary analyses	PERMANOVA (the test it qualifies).
Reporting recommendations	Report the dispersion F and P alongside every PERMANOVA.
Take-home message	A PERMANOVA is only interpretable once dispersion is shown to be homogeneous.

9.4. Pairwise PERMANOVA

Pairwise PERMANOVA is a permutation-based multivariate statistical test that compares community composition between every pair of experimental groups. It is used as a post-hoc analysis following a significant overall PERMANOVA to determine which specific groups differ from one another. Because multiple pairwise comparisons are performed, P-values are adjusted to control the false discovery rate. It requires at least three groups to be meaningful.

Table 38. Pairwise PERMANOVA plot

Output file	Pairwise_PERMANOVA_heatmap.png
What it shows	A symmetric matrix of all treatment pairs, with the BH-adjusted P value and R2 printed in each cell. Colour encodes the adjusted P value.
How it is computed	adonis2 run independently on each two-group subset, with BH correction applied across the resulting P values.
How to read it	It identifies which specific pairs differ, once the global PERMANOVA is significant. Darker cells indicate stronger separation.
What it does not prove	It requires at least three groups; with two, an informative placeholder is drawn instead, which is not an error. Each pairwise test is recomputed on its own subset, so the R2 values are not directly comparable with the global R2.

Warning signs	Uniformly high adjusted P values combined with a significant global PERMANOVA indicate that the global effect is carried by the joint structure rather than by any specific pair, which is typical of small sample sizes after BH correction.
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Table 39. Pairwise PERMANOVA quick reference

Pairwise PERMANOVA	
Biological question	Which specific pairs of treatments differ?
When to use it	As a post-hoc step after a significant global PERMANOVA, with three or more groups.
Why Triple A implements it this way	adonis2 per two-group subset with BH correction across pairs to control the false discovery rate.
Typical interpretation (example)	Control differed from both treated groups (BH-adjusted $P < 0.05$), while the treated groups did not differ from each other.
Common mistakes	Comparing pairwise R-squared with the global one (recomputed on subsets); running it without a significant global test.
Complementary analyses	Global PERMANOVA; beta dispersion.
Reporting recommendations	Report each pair's R-squared and BH-adjusted P.
Take-home message	Localises a global effect to specific pairs, after the global test is significant.

10. Community discrimination

Biological question

Can the amplicon abundance profile distinguish or predict the experimental group assigned to each sample?

10.1. Choosing parameters

PLS-DA and sPLS-DA contain parameters that strongly influence how the model is built:

- Number of components:
- Number of cross-validation folds
- Number of repeated validations (sPLS-DA only).

10.1.1. Components

Components are new variables that summarize the information contained in all taxa. In general, only $k - 1$ components (where k is the number of experimental groups) contain useful information for classification. Two components are recommended for most analyses because they provide the two-dimensional score plot. The value is additionally capped by the number of samples minus one, by the number of taxa, and by a limit derived from the cross-validation scheme itself.

Table 40. Guide for components election

Declared_groups	Components_carrying_information	Recommended
2	1	2 — the second component only provides the plane the plot is drawn on
3	2	2
4 or more	$k - 1$	2 to visualise; raise only with samples per group

10.1.2. Cross-validation folds

Cross-validation estimates how well the model would perform on independent samples by repeatedly training and testing the classifier on different subsets of the data. The number of folds cannot exceed the number of samples in the smallest experimental group. As a practical guideline:

Table 41. Guide for folds election

Biological replicates per group	Recommended folds
3 - 4	3
≥ 5	5

10.1.3. Repeated validation

For sPLS-DA, the entire cross-validation procedure is repeated several times using different random sample partitions. Repeating the validation reduces the influence of a single random split and provides a more robust estimate of model performance and feature stability.

More repetitions improve reliability but increase computation time. The default value is appropriate for most datasets, while larger numbers are recommended only when highly stable biomarker selection is required. Cross-validation estimates how well the model would perform on independent samples by repeatedly training

10.1.4. Candidate taxa per component.

Tuning tries each candidate number and keeps the one with the lowest cross-validated error, so the list is a search space rather than a setting. The default of 5, 10 and 20 suits most studies. Consider adding larger values only if the optimal model consistently selects the highest candidate tested, indicating that additional discriminative taxa may improve classification.

10.2. PLS-DA (partial least squares discriminant analysis)

PLS-DA is a supervised multivariate classification method that identifies combinations of variables (components) that maximize the separation between predefined experimental groups. It is particularly suitable for high-dimensional datasets, such as microbiome abundance tables, where the number of variables greatly exceeds the number of samples. In addition to visualizing group separation, PLS-DA can identify taxa that contributes most to sample discrimination.

The supervised nature of the method

PLS-DA deliberately constructs the axes that BEST SEPARATE the known groups. Applied to entirely random data it will still produce a figure with the groups separated.

Consequently, the PLS-DA scatter plot provides no evidence on its own. The only informative quantities are the cross-validated performance and the permutation test, both reported in the Summary sheet of PLS_DA_summary.xlsx.

10.2.1. Reading R2Y and Q2

The Summary sheet of PLS_DA_summary.xlsx reports both. **R2Y** measures how much of the class structure the model reproduces on its own training data; **Q2** measures how much it reproduces on samples it has not seen. A high R2Y with a low Q2 is the signature of overfitting, and it is the expected pattern with few samples and many taxa.

The value to be cited as predictive ability is Q², not R²_Y. As indicative guidance for omics data, a Q² above 0.4 indicates a model with reasonable predictive ability; below zero the model predicts worse than the mean and must not be interpreted, however well separated the point cloud appears.

10.2.2. Permutation test

The Summary sheet includes a permutation P value obtained by relabelling the treatments at random and refitting the model the configured number of times. This is the key check for PLS-DA: it answers whether the accuracy obtained is better than that achievable with random labels. A non-significant permutation P value invalidates the model however high the accuracy appears. With 99 repetitions the minimum attainable P value is 0.01.

Table 42. PLS-DA score plot

Output file	PLS_DA.png
What it shows	Samples projected onto the latent components of the supervised model. The title includes the cross-validated accuracy.
How it is computed	pls::plsr with k-fold cross-validation. Taxa are centred and scaled to unit variance before fitting (autoscaling, standard in PLS-DA); classes are dummy coded.
How to read it	The first thing to read is the percentage in the title, not the cloud of points. A cross-validated accuracy close to chance (50% with two groups, 33% with three) invalidates any interpretation of the figure, however well separated it appears.
What it does not prove	Visual separation is not evidence. Neither are the ellipses: they are computed on coordinates that have already been optimized for separation.
Warning signs	The number of components is limited by the rank of the data and by what cross-validation can support, not by the number of groups: with two treatments two components are fitted and the plot is two-dimensional. If a single component still appears, the sample size or the number of taxa does not allow more.

Table 43. PLS-DA confusion matrix

Output file	PLS_DA_confusion_matrix.png
What it shows	An observed x predicted contingency table, with the count printed in each cell.
How it is computed	Predictions obtained from the held-out cross-validation segments, not from the fit on the training data. It is therefore an honest estimate.
How to read it	The diagonal contains the correct assignments. Off-diagonal cells state which groups the model confuses, which is biological information: if an intermediate treatment is confused with the control rather than with the extreme, there is a response gradient.
What it does not prove	A clean diagonal does not amount to significance. With few samples the accuracy carries enormous uncertainty: eleven correct out of twelve is compatible with a very wide confidence interval.
Warning signs	Cross-validation segments are formed at random, without stratification by group. With few replicates per treatment a segment may contain no representative of a class, which degrades the estimates. Reducing cv_folds helps in those cases.

Table 44. Variable importance in projection (VIP)

Output file	PLS_DA_VIP.png
What it shows	The twenty taxa with the highest VIP score, as horizontal bars, with a dashed line at VIP = 1.
How it is computed	The standard VIP formula from the model X weights and the Y variance explained by each component.
How to read it	The conventional VIP > 1 threshold is used as a screening criterion: these are the taxa contributing most to the discrimination. The reference line is drawn for exactly that purpose.
What it does not prove	VIP > 1 is neither a significance test nor a P value. It is a rule of thumb with no associated null distribution. A taxon with a high VIP in a model whose cross-validation is poor means nothing.
Warning signs	Always check the validated performance before reading this figure. A VIP ranking extracted from an overfitted model is a list of taxa selected by noise, and this is a frequent error in the literature.

Table 45. PLS-DA cross-validation performance

Output file	PLS_DA_summary.xlsx (Summary sheet).
What it contains	Samples, Classes, Components, CV folds, Cross-validated accuracy, Balanced accuracy, R2Y, Q2, Permutation repetitions, Permutation p-value.
How it is computed	k-fold CV of pls::plsr plus a permutation test on relabelled treatments.
How to read it	Cite Q2 and the permutation P, not the score plot.
What it does not prove	High R2Y with low Q2 is overfitting; Q2 < 0 or non-significant P invalidates the model.

Table 46. PLS-DA taxon loadings

Output file	PLS_DA_summary.xlsx (Loadings sheet).
What it contains	Loading of each taxon on each component.
How it is computed	Component weights of the fitted model.
How to read it	Sign and magnitude give the direction and strength each taxon contributes; complements the VIP figure.
What it does not prove	Loadings from a poorly validated model describe noise.

Table 47. PLS-DA quick reference

PLS-DA	
Biological question	Can the abundance profile predict a sample's treatment group?
When to use it	For supervised discrimination when taxa greatly outnumber samples.
Why Triple A implements it this way	pls::plsr with autoscaling and k-fold CV; because it is supervised, only cross-validated performance and the permutation test are evidence.
Typical interpretation (example)	PLS-DA discriminated the groups with cross-validated accuracy 83% (Q2 = 0.55, permutation P = 0.01).
Common mistakes	Interpreting the score plot or ellipses as evidence: a supervised model separates even random data.
Complementary analyses	Q2 and permutation test (Summary sheet); VIP (drivers).
Reporting recommendations	Report Q2, cross-validated accuracy and the permutation P; not R2Y, not the plot.
Take-home message	The plot proves nothing; only validated performance does.

10.3. sPLS-DA (sparse PLS-DA)

sPLS-DA is a supervised multivariate classification method that extends PLS-DA by simultaneously performing sample classification and variable selection. Instead of using all taxa, sPLS-DA automatically selects a reduced subset of the most informative taxa (a microbial signature) for each component, producing simpler and more interpretable models while reducing noise. The optimal number of taxa is determined by repeated stratified cross-validation implemented in the mixOmics package. Model performance should be assessed using the cross-validated classification error and the stability of the selected taxa across repeated validations rather than by the apparent simplicity of the resulting signature.

Table 48. sPLS-DA score plot

Output file	sPLS_DA.png
What it shows	Samples projected onto the sparse components. The title reports the validated balanced error rate (BER).
How it is computed	mixOmics::splsd with keepX tuned by tune.splsd (repeated M-fold validation, centroid distance, BER criterion) and evaluated with mixOmics::perf.

How to read it	The quality indicator here is BER, not accuracy. BER is the mean of the per-class error rates, so it is not misled by unbalanced classes. A BER near zero indicates excellent discrimination; a BER near $(1 - 1/\text{number of classes})$ is equivalent to chance.
What it does not prove	The same caveat as PLS-DA: it is a supervised method and will separate random groups.
Warning signs	If the automatic keepX tuning selects the maximum number of candidate taxa on every component, the model is not genuinely sparse: widen the candidate list or reconsider whether there is sufficient signal.

Table 49. sPLS-DA confusion matrix

Output file	sPLS_DA_confusion_matrix.png
What it shows	An observed x predicted table for the selected signature.
How it is computed	IMPORTANT: unlike the PLS-DA matrix, this one is built from predictions on the SAME training data, not on held-out segments. The application states this correctly by labelling the corresponding accuracy as "descriptive only" in the Summary sheet.
How to read it	It is useful for seeing which groups are confused, but any accuracy derived from it is optimistically biased by construction.
What it does not prove	It must not be reported as model performance. The value to cite is the validated BER from the Summary sheet.
Warning signs	A perfect diagonal here is normal and means nothing: the model is being evaluated on the data it was fitted to.

Table 50. Selected microbial signature

Output file	sPLS_DA_selected_features.png
What it shows	The thirty selected taxa with the largest absolute loading, coloured by the component on which they were selected.
How it is computed	mixOmics::selectVar by component. The Component_recurrence sheet of the workbook counts on how many components each taxon appears.
How to read it	This is the most translationally valuable output of the module: a reduced set of candidate biomarker taxa. Taxa appearing on several components are the most consistent candidates.
What it does not prove	Selection is not a per-taxon statistical test. A selected taxon is not "significantly associated" with the treatment; it is contributing to a multivariate model.
Warning signs	The Component_recurrence table measures recurrence ACROSS COMPONENTS, not selection stability across cross-validation resamples, which is what stability usually denotes in the mixOmics literature. The table name is accurate; the distinction should be preserved when writing.

Table 51. sPLS-DA cross-validation performance

Output file	sPLS_DA_summary.xlsx (Summary sheet).
What it contains	Samples, Classes, Components, CV folds, CV repeats, Selected taxa per component, Fitted accuracy (descriptive only), Balanced fitted accuracy (descriptive only), Cross-validated BER.
How it is computed	Repeated stratified CV (tune.splsda, centroid distance, BER), evaluated with mixOmics::perf.
How to read it	Cite the Cross-validated BER (robust to unbalanced classes).
What it does not prove	The fitted accuracies are descriptive only and must not be reported as performance.

Table 52. sPLS-DA feature-selection stability

Output file	sPLS_DA_summary.xlsx (Component_recurrence sheet).
What it contains	Each selected taxon with the number of components on which it was selected.
How it is computed	mixOmics::selectVar counted across components.
How to read it	Taxa on several components are the most consistent candidates.
What it does not prove	Recurrence across components is not stability across CV resamples.

Table 53. sPLS-DA quick reference

sPLS-DA	
Biological question	Which minimal set of taxa discriminates the groups?
When to use it	When a reduced, interpretable biomarker signature is the goal.
Why Triple A implements it this way	mixOmics splsda with keepX tuned by repeated CV under the BER criterion, which is robust to unbalanced classes.
Typical interpretation (example)	A signature of 15 taxa discriminated the groups (BER = 0.08 by repeated cross-validation).
Common mistakes	Reporting the training-data confusion matrix (descriptive only) as performance; equating selection with per-taxon significance.
Complementary analyses	Cross-validated BER (Summary); component recurrence (consistency).
Reporting recommendations	Report the validated BER, the number of components and taxa selected, and the signature.
Take-home message	Cite the validated BER and the signature; selection is not a significance test.

11. Environmental analysis

Biological question

Which environmental gradients are associated with the community, how much variation do they explain, and what remains after controlling for confounders?

11.1. Environmental fit (envfit)

Fits environmental variables onto an existing unconstrained ordination as vectors, or as centroids for factors, and tests each by permutation. It is the lightest way to ask which measured variables align with the patterns already visible in a PCA, PCoA or NMDS. It does not constrain the ordination, so it describes association rather than explained variation.

Table 54. Envfit plot

Output file	envfit_envfit.png
What it shows	An UNCONSTRAINED ordination (Hellinger PCA) onto which the best-fitting environmental variables are superimposed.
How it is computed	vegan::rda without constraints followed by vegan::envfit with permutations. It accepts both continuous variables (arrows) and factors (centroids).
How to read it	It differs from RDA in one key conceptual respect: here the ordination is computed WITHOUT reference to the environment, and the variables are projected afterwards. It answers "which variables align with the structure that exists naturally?", whereas RDA answers "how much structure can these variables explain?". The Vectors and Factor_centroids tables report the R2 and P value of each variable.
What it does not prove	The envfit R2 measures fit to the two displayed axes only, not to the complete community.
Warning signs	P values are not corrected for multiple comparisons. With many environmental variables, apply the correction manually to the P_value column before reporting.

Table 55. Envfit quick reference

envfit	
Biological question	Which measured variables align with the patterns already visible in an unconstrained ordination?
When to use it	The lightest way to relate environment to an existing PCA/PCoA/NMDS without constraining it.

envfit	
Why Triple A implements it this way	vegan::envfit projects variables as vectors/centroids onto an unconstrained ordination and tests each by permutation, so it describes association rather than explained variation.
Typical interpretation (example)	Temperature was significantly associated with the ordination (envfit R2 = 0.70, P = 0.001).
Common mistakes	Reading fit to two axes as fit to the whole community; forgetting the P values are uncorrected.
Complementary analyses	RDA (for explained variation); manual BH correction of the P column.
Reporting recommendations	Report each variable's R2 and P, and that P is uncorrected (or correct it).
Take-home message	Describes which variables align with existing structure, not how much they explain.

11.2. RDA (redundancy analysis)

A constrained ordination. Instead of seeking the axes of maximum variation as PCA does, it seeks the axes of maximum variation explainable by the supplied environmental variables. It is the appropriate tool for relating composition to measured gradients.

The implementation is a tb-RDA: RDA on the Hellinger-transformed matrix, which is the standard recommended practice for compositional data (Legendre and Gallagher, 2001). Environmental variables are standardized before fitting, so arrow lengths are comparable between variables with different units.

Table 56. RDA ordination plot (biplot)

Output file	RDA.png
What it shows	Samples as points coloured by treatment, and environmental variables as arrows from the origin. The subtitle reports the constrained variance, the global permutation P value and the number of retained predictors.
How it is computed	vegan::rda on the Hellinger matrix with standardized environmental variables. Arrows are the biplot scores. Their length is rescaled only so that they fit within the panel; the analytical magnitudes exported to the workbook are unaltered.
How to read it	The arrow direction is the direction of increase of that variable. Samples lying in the direction of an arrow have high values of it. Arrows pointing the same way indicate correlated variables; opposing arrows indicate negative correlation; perpendicular arrows indicate independence. Length reflects the strength of association with the displayed axes.
What it does not prove	The rescaled length is NOT comparable between different figures. Above all, visual alignment between an arrow and a group of samples is descriptive. Significance is given exclusively by the permutation tests in the Term_tests sheet.
Warning signs	Always consult the Predictor selection sheet. If four variables were selected but only two arrows appear, the application excluded the other two and the reason is recorded there: constant, linearly dependent, or dropped to preserve residual degrees of freedom. Consult the VIF sheet as well: values above 10 indicate high collinearity and make the individual per-variable coefficients uninterpretable.

Table 57. RDA constrained variance per axis

Output file	RDA_variance.png
What it shows	The percentage of constrained variance corresponding to each RDA axis, up to a maximum of ten axes.
How it is computed	100 x axis eigenvalue / sum of the constrained eigenvalues.
How to read it	It shows how the portion of variation that the environmental variables do explain is distributed across axes. If the first axis concentrates almost all of it, a single environmental gradient dominates the community response.
What it does not prove	THESE PERCENTAGES ARE RELATIVE TO THE CONSTRAINED PART, NOT TO THE TOTAL. By construction they sum to 100%. A first axis with 85% here may represent 19% of the total variation if the model explains only 22%.

Warning signs	The figure to report as "how much the environment explains" is the adjusted R ² in the Summary sheet, not these percentages. The row "Constrained variance (% of total inertia)" in that same sheet gives the fraction of TOTAL variation captured by the model and equals 100 x R ² ; it is also the value shown in the subtitle of RDA.png.
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Table 58. RDA permutation test

Output file	RDA_summary.xlsx (Overall_test, Term_tests and Axis_tests sheet).
What it contains	Permutation significance for the model, each variable and each axis: Df, Variance, F, Pr(>F).
How it is computed	Permutation tests on the vegan::rda model.
How to read it	Term_tests holds per-variable significance; check Predictor_selection and VIF too.
What it does not prove	Per-variable coefficients are uninterpretable when VIF > 10.

Table 59. RDA explained variance

Output file	RDA_summary.xlsx (explained_variance sheet).
What it contains	Each constrained axis with its % of constrained variance and cumulative %.
How it is computed	100 x axis eigenvalue / sum of constrained eigenvalues.
How to read it	Shows how the explained variation is distributed across axes.
What it does not prove	Relative to the constrained part (sums to 100%), NOT of total; report adjusted R-squared instead.

Table 60. RDA quick reference

RDA	
Biological question	Which environmental variables explain community composition, and how much?
When to use it	To relate composition to measured continuous gradients (constrained ordination).
Why Triple A implements it this way	tb-RDA: RDA on the Hellinger matrix with standardized predictors, the standard for compositional data (Legendre & Gallagher 2001), so arrows are comparable across units.
Typical interpretation (example)	Environmental variables explained 22% of compositional variation (adjusted R ² = 0.18); temperature was the strongest predictor (P = 0.006).
Common mistakes	Reading arrow-sample alignment as significance; citing per-axis constrained % as total variation; ignoring collinearity.
Complementary analyses	VIF and adjusted R-squared (Summary); Term_tests (per-variable P); Predictor_selection.
Reporting recommendations	Report adjusted R-squared, the constrained % of total inertia, per-variable P, and confirm VIF < 10.
Take-home message	Report adjusted R-squared and per-variable permutation P, not the biplot geometry.

11.3. Partial constrained ordination (partial RDA / dbRDA)

RDA that first removes the effect of declared covariates, a batch, a site, a sampling date, and then asks what the variables of interest explain in the remainder. This is the tool for separating a real environmental effect from a confounded study design.

Table 61. partial RDA ordination plots

Output file	<Partial_RDA dbRDA Partial_dbRDA>_ordination.png
What it shows	A constrained ordination from which the effects of a set of conditioning variables (covariates) have first been removed.
How it is computed	vegan::rda with a Condition() term when no distance is specified, or vegan::capscale with an additive correction when one is (distance-based RDA). The figure title states which of the four variants was run.

How to read it	This is the tool for isolating the effect of interest from a known confounder, for example evaluating the effect of pH after accounting for sampling site. The Model variables sheet records which variables were explanatory and which were conditioning.
What it does not prove	Accounting for a covariate is not equivalent to controlling it experimentally. If the covariate is correlated with the treatment, the shared effect disappears from the model and the result may underestimate the true effect.
Warning signs	The Percent column of the Variance sheet is computed over ALL eigenvalues, constrained and unconstrained. It is not directly comparable with the percentages in RDA variance.png of the RDA module.

Table 62. *partial RDA quick reference*

Partial RDA / dbRDA	
Biological question	What do the variables of interest explain after removing a known confounder?
When to use it	When a batch, site or sampling date confounds the effect you care about.
Why Triple A implements it this way	vegan::rda with a Condition() term (or capscale for distance-based) to partial out covariates before testing the predictors of interest.
Typical interpretation (example)	After accounting for sampling site, pH still explained a significant fraction of composition ($P = 0.01$).
Common mistakes	Treating statistical adjustment as experimental control; underestimating an effect confounded with the covariate.
Complementary analyses	Variance partitioning (how confounded the sets are); plain RDA.
Reporting recommendations	Report which variables were explanatory vs conditioning, and the conditioned R-squared and P.
Take-home message	Isolates an effect from a known confounder; adjustment is not control.

11.4. Variance partitioning

Decomposes the explained variation between sets of predictors (typically environmental variables against experimental factors) into the part unique to each set and the part they share. The shared fraction is often the most informative result, because it measures how confounded the two sets are with each other.

Table 63. *Variance partitioning fractions plot*

Output file	Variance_partitioning_fractions.png
What it shows	Bars with the adjusted R2 of each fraction: explained by the environmental variables alone, by the experimental factors alone, shared between both, and residual.
How it is computed	vegan::varpart on the Hellinger matrix with two sets of predictors.
How to read it	It separates how much of the observed effect is attributable to the experimental design, how much to the measured environment, and how much is indistinguishable between the two. The shared fraction is usually the most informative: if it is large, treatment and environmental gradient are confounded and cannot be separated with these data.
What it does not prove	The shared fraction is not an interaction effect. It is variance that both sets explain equally well and that this design cannot attribute.
Warning signs	Fractions with a NEGATIVE adjusted R2 are normal and should be interpreted as zero: they mean that the predictor set explains less than a set of random variables of the same size would.

Table 64. *Variance partitioning quick reference*

Variance partitioning	
Biological question	How much variation is unique to the environment, unique to the design, and shared?
When to use it	When environmental gradients and experimental factors may be confounded.

Variance partitioning	
Why Triple A implements it this way	vegan::varpart on the Hellinger matrix; the shared fraction quantifies confounding directly.
Typical interpretation (example)	Treatment and pH shared 30% of explained variation, indicating the two are strongly confounded in this design.
Common mistakes	Reading the shared fraction as an interaction; reporting a negative adjusted R-squared as anything but zero.
Complementary analyses	Partial RDA (test of the unique fractions); RDA.
Reporting recommendations	Report each fraction's adjusted R-squared and interpret negatives as zero.
Take-home message	The shared fraction measures confounding, not interaction.

12. Taxon associations

Biological question

Which taxa are more abundant in one group, or change along environmental gradients after adjustment for other variables?

12.1. Analysis of compositions of microbiomes with bias correction - 2 (ANCOM-BC2)

This tests individual taxa for differential abundance while explicitly modelling the compositional nature of the data and correcting for sampling bias. It supports covariates and multi-group contrasts. Because it models sequencing depth, it must receive genuine counts, a table declared as counts but holding rounded percentages loses exactly the information the method depends on, which is why the abundance scale is validated at import.

Table 65. ANCOM-BC2 volcano plot

Output file	ANCOM_BC2_volcano.png
What it shows	Bias-corrected log fold change against -log10 of the adjusted P value, with one panel per model term when there is more than one.
How it is computed	ANCOMBC::ancombc2 with BH correction, structural-zero detection and pairwise contrasts. Points marked as significant are those ANCOM-BC2 itself declares differential.
How to read it	ANCOM-BC2 is a compositional method: it models and corrects the sampling bias introduced by the fact that abundances sum to a constant. It is more appropriate than Wilcoxon for amplicon data and accepts covariates, which Wilcoxon does not.
What it does not prove	The log fold change is on the model's bias-corrected scale; it is not the log2FC of the differential-abundance volcano plot and must not be compared numerically with it.
Warning signs	ANCOM-BC2 exploits the real sequencing depth, so importing the table declared as counts is the best option when counts are available. If only proportions or percentages exist, the application constructs counts per million: the analysis runs, but bias correction operates on identical fictitious depths and the results are less reliable.

Table 66. ANCOM-BC2 quick reference

ANCOM-BC2	
Biological question	Which individual taxa are differentially abundant, accounting for compositionality and sampling bias?
When to use it	The preferred per-taxon test for amplicon data, especially with covariates or multi-group designs.
Why Triple A implements it this way	ANCOMBC::ancombc2 models sequencing depth and corrects sampling bias, so it needs genuine counts; more appropriate than Wilcoxon for compositional data.

ANCOM-BC2	
Typical interpretation (example)	ANCOM-BC2 identified 8 genera as differentially abundant between treatments (BH-adjusted $P < 0.05$).
Common mistakes	Feeding it rounded percentages (loses depth); comparing its bias-corrected log fold change numerically with the Wilcoxon volcano.
Complementary analyses	MaAsLin2 (multivariable cross-check); pairwise differential abundance.
Reporting recommendations	Report the bias-corrected log fold change, adjusted P, and that input was genuine counts.
Take-home message	The compositional test of choice, but only on real counts.

12.2. Multivariable association between taxa and metadata (MaAsLin2)

Fits a model per taxon with several predictors at once, so a taxon's association with one variable is estimated while adjusting for the others. It handles both categorical and continuous predictors, which makes it the natural choice when the design is not a simple group comparison. Results are reported as coefficients with adjusted P-values.

Table 67. MaAsLin2 forest plot

Output file	MaAsLin2_forest.png
What it shows	The twenty-five most significant associations: model coefficient with its approximate 95% confidence interval, ordered by magnitude and coloured according to whether they pass the q threshold.
How it is computed	Maaslin2 with a linear model on log-transformed, standardized relative abundance, with BH correction. It accepts fixed and random effects.
How to read it	Each row is a taxon-variable pair. The coefficient gives the direction and magnitude of the association; intervals that do not cross the vertical zero line correspond to robust associations. Random effects allow repeated measures or block structure to be handled correctly, which is their principal advantage.
What it does not prove	The intervals are approximate (coefficient plus or minus 1.96 times the standard error), not the exact model intervals.
Warning signs	With factors of more than two levels, MaAsLin2 requires a reference level; Triple A selects the first in alphabetical order automatically. If the control is not alphabetically first, the signs of the coefficients will be inverted relative to expectation. Check the Fixed_effects sheet.

Table 68. MaAsLin2 quick reference

MaAsLin2	
Biological question	Which taxa are associated with a variable while adjusting for the others?
When to use it	When the design is not a simple group comparison: mixed continuous/categorical predictors or repeated measures.
Why Triple A implements it this way	Maaslin2 fits a per-taxon linear model on log-transformed, standardized abundance with BH correction; random effects handle block or repeated structure.
Typical interpretation (example)	MaAsLin2 associated Genus X with pH after adjusting for temperature ($q < 0.05$).
Common mistakes	Misreading coefficient signs when the reference level is not the intended control (Triple A picks the first alphabetically).
Complementary analyses	ANCOM-BC2 (compositional cross-check); the Fixed_effects sheet for the reference level.
Reporting recommendations	Report the coefficient, its adjusted P (q) and the reference level.
Take-home message	The multivariable choice; verify the reference level before reading signs.

12.3. Pairwise differential abundance

A direct two-group comparison per taxon, filtered by prevalence and mean abundance before testing to reduce the multiple-testing burden. Results are presented as a volcano plot (effect size against significance), an MA plot (effect size against mean abundance) and a QQ plot of the P-values, which is the quickest way to see whether the test behaved as expected. A taxon is highlighted only when it passes both the adjusted-P threshold and the fold-change threshold

Table 69. *Volcano plot*

Output file	Volcano_<comparison>.png
What it shows	Each point is a taxon. X axis: log2 of the abundance change (positive means more abundant in the second treatment of the title). Y axis: -log10 of the adjusted P value.
How it is computed	$\log_2FC = \log_2((\text{mean}_B + \text{pseudocount}) / (\text{mean}_A + \text{pseudocount}))$ on relative abundances. The pseudocount (1e-6 by default) prevents division by zero. Dashed vertical lines mark the $ \log_2FC $ threshold and the horizontal line the alpha threshold.
How to read it	The taxa of interest lie at the top and towards the sides: high significance and large effect. Only points simultaneously above the horizontal line and outside the vertical lines are colour-coded as significant.
What it does not prove	A large log2FC in a rare taxon may be an artefact of the pseudocount: moving from 0.001% to 0.01% is a log2FC of +3.3 with no ecological relevance. Always cross-check against the MA plot, which places each point at its abundance level.
Warning signs	Finding no point above the significance line with a small sample size is expected rather than a failure: with three or four replicates per group the minimum attainable Wilcoxon P value may not survive BH correction. If effects must be detected with few replicates, use ANCOM-BC2 or MaAsLin2.

Table 70. *MA plot*

Output file	MA_<comparison>.png
What it shows	X axis: log10 of the mean abundance of the taxon in the comparison. Y axis: log2FC. Same colour coding as the volcano plot.
How it is computed	The same data and the same significance classification as the volcano plot; only the X axis changes.
How to read it	This is the diagnostic for dependence between effect and abundance. In a healthy experiment the cloud is centred on log2FC = 0 and narrows towards the right, because abundant taxa are estimated more precisely and show less variance.
What it does not prove	It adds no statistical information relative to the volcano plot: it is the same table viewed from another angle.
Warning signs	A cloud shifted globally above or below zero indicates a compositional imbalance between groups, typically a highly dominant taxon that changes and drags the rest through the constant-sum effect. In that scenario compositional methods such as ANCOM-BC2 are more reliable than Wilcoxon.

Table 71. *P-value QQ plot*

Output file	QQ_<comparison>.png
What it shows	Observed quantiles of -log10(P) against the quantiles expected under the null hypothesis, that is, a uniform distribution. A dashed reference diagonal is drawn.
How it is computed	Uses the UNADJUSTED P values, sorted, against -log10(ppoints(n)). It is a diagnostic, not a test.
How to read it	If the points follow the diagonal, the test is well calibrated and there is little or no real signal. An upper tail that departs upwards at the end is the pattern expected when true positives are present: most taxa do not differ and a few do.
What it does not prove	It does not indicate which taxa are significant; the volcano plot serves that purpose.
Warning signs	The whole cloud above the diagonal (inflation) indicates systematically small P values: unmodelled dependence between samples, pseudoreplication, or replicates that are not independent. The whole cloud below (deflation) is typical of discrete tests with very few replicates and indicates low power.

Table 72. Pairwise differential abundance quick reference

Pairwise differential abundance	
Biological question	Which taxa change between two specific groups?
When to use it	For a direct, interpretable two-group comparison per taxon.
Why Triple A implements it this way	Prevalence and abundance filtering then a per-taxon test, shown as volcano/MA/QQ; a taxon is flagged only if it passes both the adjusted-P and fold-change thresholds.
Typical interpretation (example)	Genus X was enriched in treated samples ($\log_2FC = 2.1$, adjusted $P = 0.03$).
Common mistakes	Over-interpreting a large $\log_2FC$ in a rare taxon (pseudocount artefact); expecting hits with very few replicates.
Complementary analyses	MA plot (abundance dependence); QQ plot (calibration); ANCOM-BC2/MaAsLin2 when replicates are few.
Reporting recommendations	Report $\log_2FC$, adjusted P and mean abundance for each flagged taxon.
Take-home message	Effect size and significance together; check the MA plot before trusting a fold change.

13. Functional analysis

Biological question

Which curated biological functions are potentially represented, how abundant are their contributing taxa, and which functions differ between conditions?

13.1. Functional potential

Links each taxon to a representative NCBI RefSeq genome, searches that genome's GFF annotation for the curated marker genes of each selected function, and applies conservative rule-based logic over diagnostic, supporting and accessory gene evidence. The result is a call per taxon and function, accompanied by module completeness and a confidence tier of High, Medium, Low or Insufficient evidence. This is inferred genomic potential in a reference genome: it is not gene expression, not measured activity, and not proof that the specific strain in the sample carries the pathway.

Table 73. Figures performed with the functional potential module

Output file	<configured title>.png
What it shows	Functional categories in rows, treatments in columns, with the aggregated abundance of the taxa assigned to each category.
How it is computed	Rule-based classification using diagnostic, supporting and accessory gene evidence. The Taxon results table additionally reports module completeness and a confidence tier (High / Medium / Low / Insufficient evidence).
How to read it	ALWAYS alongside the confidence column. A category sustained by low-confidence calls is not comparable with one sustained by high-confidence calls.
What it does not prove	This is INFERRED GENOMIC POTENTIAL from a reference genome, not measured activity or gene expression. A taxon may carry the pathway and never express it under the experimental conditions. The correct wording is "potential for", never "capable of" or "performs".
Warning signs	"Insufficient evidence" is a legitimate result, not an error: it means the taxon could not be resolved to a representative genome, or that the gene evidence did not reach the threshold. Manganese, selenium and anaerobic methane oxidation are deliberately excluded because their marker genes cannot be reliably identified by name.

Figure	What it adds and how to read it
..._heatmap.png	Pathway x treatment with the mean abundance ($\pm$ standard deviation across replicates) printed in each cell. The valid comparison is within a row.
..._barplot.png	Bars grouped by treatment, one series per pathway, with error bars of one standard deviation. Useful for comparing pathways with each other within a treatment.
..._dotplot.png	Pathway x treatment where size and colour encode abundance. A compact format, preferable when there are many pathways.
..._top_contributors.png	The taxa contributing most to each pathway, one panel per pathway. It answers which specific organisms sustain the functional signal.

Table 74. Functional potential quick reference

Functional potential	
Biological question	Which curated functions could the community's taxa carry?
When to use it	To infer metabolic capabilities from reference genomes, once taxonomy is resolved.
Why Triple A implements it this way	It links each taxon to a RefSeq genome and applies conservative rule-based logic over marker genes, reporting a confidence tier, because genomic potential is an inference, not a measurement.
Typical interpretation (example)	Nitrogen-fixation potential was inferred with high confidence in four of the dominant genera.
Common mistakes	Wording potential as activity ('performs', 'capable of') instead of 'potential for'; ignoring the confidence tier.
Complementary analyses	Functional abundance (weights by abundance); always the confidence column.
Reporting recommendations	Report the confidence tier and that results are inferred genomic potential from reference genomes.
Take-home message	Inferred potential in a reference genome, never measured activity.

Pathways overlap taxonomically

The same taxon may be assigned to several pathways simultaneously. The abundances of different pathways are NOT additive and do not sum to 100%. The figure itself states this in the subtitle.

When two pathways share their principal contributors, their joint change between treatments is a single biological phenomenon counted twice, not two independent findings. Always check this against the top-contributors figure.

13.2. Functional abundance

Weights the functional calls by how abundant the contributing taxa actually are, turning a presence-oriented result into a quantitative profile of potential metabolic pathways. It also identifies the top contributing taxa per pathway, which is usually the more actionable output: it says which organisms are responsible for a pathway's signal.

Table 75. Functional abundance quick reference

Functional abundance	
Biological question	How abundant are the taxa that carry each function?
When to use it	To turn presence/absence functional calls into a quantitative pathway profile.
Why Triple A implements it this way	It weights functional calls by contributor abundance and lists the top contributors, usually the more actionable output.
Typical interpretation (example)	The methanogenesis pathway was most abundant in the anaerobic treatment, driven mainly by Methanosarcina.

Functional abundance	
Common mistakes	Summing pathway abundances (they overlap taxonomically and are not additive); counting two pathways with the same contributors as two findings.
Complementary analyses	Top-contributors figure (which organisms); functional potential (confidence).
Reporting recommendations	Report per-pathway abundance +/- SD and the main contributing taxa.
Take-home message	A quantitative potential profile; pathways overlap, so they do not sum.

13.3. Differential functions

Compares the inferred functional profiles between declared experimental groups, asking which capabilities are differentially represented rather than which taxa are. It runs on the pathway-by-sample matrix that the functional abundance analysis already produces, so no abundance is recomputed. Each pathway is tested with a Wilcoxon rank-sum test for every pair of treatments, and the P values are Benjamini-Hochberg adjusted within each comparison. The non-parametric test is deliberate: pathway abundances inherit the skew of the taxon abundances they are summed from, so normality cannot be assumed. A pathway that is constant across every sample is reported as untested rather than given a P value. It inherits every limitation of the underlying inference: a difference in potential is not a difference in activity.

Table 76. *Differential functions workbook*

Output file	The differential-functions workbook.
What it contains	Per pathway and treatment pair: Mean_A, Mean_B, Difference, log2FC, P_value, Tested, Adjusted_P, Significance.
How it is computed	Wilcoxon per pair of treatments, BH adjusted within each comparison.
How to read it	Which inferred capabilities differ between groups; constant pathways are reported untested.
What it does not prove	A difference in potential is not a difference in activity.

Table 77. *Differential function quick reference*

Differential functions	
Biological question	Which inferred capabilities differ between groups?
When to use it	After functional abundance, to compare pathway profiles rather than taxa.
Why Triple A implements it this way	Wilcoxon per pair of treatments with BH within each comparison; non-parametric because pathway abundances inherit taxon-abundance skew.
Typical interpretation (example)	Sulfate-reduction potential was higher in sediment than in water samples (BH-adjusted P = 0.02).
Common mistakes	Reading a difference in potential as a difference in activity; interpreting untested constant pathways.
Complementary analyses	Functional abundance (the input); the top-contributors figure.
Reporting recommendations	Report the pathway, the higher group, log2FC and adjusted P.
Take-home message	Differences in potential between groups, still not activity.

13.4. Functional enrichment

Asks whether a functional theme, rather than a handful of isolated taxa, characterises a condition. It crosses the per-taxon functional calls with the taxa that the differential abundance analysis flagged as significant, and tests each function with Fisher's exact test, Benjamini-Hochberg adjusted across functions. The definition of the background is what makes or breaks this analysis, so it is worth stating plainly: the background is every taxon that received a

functional call, not every taxon in the dataset. A taxon without a resolved reference genome could never have been called positive, and counting it would inflate the apparent enrichment. Read the overlap against the expected overlap: an odds ratio above one resting on two or three taxa is fragile evidence however small the P value.

Table 78. Functional enrichment workbook

Output file	The functional-enrichment workbook.
What it contains	Per function: Significant_taxa, Overlap, Expected, Odds_ratio, P_value, Adjusted_P, Enriched.
How it is computed	Fisher's exact test per function, BH adjusted; background = taxa with a functional call.
How to read it	Read observed Overlap against Expected.
What it does not prove	An odds ratio resting on 2-3 taxa is fragile however small the P.

Table 79. Functional enrichment quick reference

Functional enrichment	
Biological question	Does a functional theme characterise the differentially abundant taxa?
When to use it	To ask whether a capability, not just isolated taxa, marks a condition.
Why Triple A implements it this way	Fisher's exact test per function with BH, using taxa that received a functional call as the background, because untested taxa could never be positive and would inflate enrichment.
Typical interpretation (example)	Chemolithotrophy was over-represented among taxa enriched in the treated group (odds ratio 4.2, adjusted P = 0.01).
Common mistakes	Trusting an odds ratio resting on two or three taxa; using all taxa as the background.
Complementary analyses	Differential abundance (defines the significant set); functional potential.
Reporting recommendations	Report the odds ratio, observed vs expected overlap and adjusted P.
Take-home message	Read overlap against expectation; small-overlap enrichment is fragile.

14. Glossary and references

Term	Operational definition in Triple A
Relative abundance	The percentage a taxon represents within its own sample. The entire application works on this scale.
Hellinger transformation	The square root of relative abundance. It makes the Euclidean distance appropriate for compositional data.
Bray-Curtis	A dissimilarity that weights abundances. It is the default metric.
Jaccard (binary)	A presence/absence dissimilarity. It gives far more weight to rare taxa.
Stress (NMDS)	Mismatch between the original distances and those represented. Below 0.10 good; 0.10 to 0.20 acceptable; above 0.20 poor.
PERMANOVA R ²	Fraction of compositional variation attributable to the model term. It is the effect size.
Beta dispersion	Variability within each group. Its homogeneity is an assumption of PERMANOVA.
ANOSIM R	A rank-based contrast between inter- and intra-group distances, from -1 to 1.
log ₂ FC	Base-2 logarithm of the ratio of mean abundances between two treatments. A value of +1 means twofold.
Adjusted P (BH)	P value corrected for multiple comparisons with Benjamini-Hochberg. This is the one to use.
VIP	Importance of a variable in the PLS-DA projection. VIP > 1 is a screening criterion, not a test.
BER	Balanced error rate: the mean of the per-class error rates. Robust to unbalanced classes.
Q ²	Validated predictive ability of a PLS model. It is the value to cite, ahead of R ² _Y . A Q ² below zero means the model predicts worse than the mean.
VIF	Variance inflation factor. Above 10, collinearity between predictors is problematic.
Constrained variance	In RDA, the portion of compositional variation explainable by the environmental variables.
Functional potential	Presence of marker genes in a representative reference genome. It is not measured activity.

14.1. Methodological references

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